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SURVEY

Drone-Based Delivery Systems: A Survey on Route Planning

GIULIO ATTENNI^{ID}, (Graduate Student Member, IEEE), VIVIANA ARRIGONI^{ID}, (Member, IEEE), NOVELLA BARTOLINI^{ID}, (Senior Member, IEEE), AND GAIA MASELLI^{ID}, (Member, IEEE)

Computer Science Department, Sapienza University of Rome, 00185 Rome, Italy

Corresponding author: Giulio Attenni (giulio.atteni@uniroma1.it)

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ABSTRACT The ever-increasing demand for urban delivery services significantly exacerbates the traffic conditions on our already overly congested urban streets. The problem is even worsened by the dearth of appropriate parking facilities available for efficient offloading operations. As a result, there has been a surge of interest in exploring eco-friendly alternatives to traditional truck-based logistics. Drone technology has significant potential to enable widespread home delivery services and reduce the reliance on trucks within urban areas. Both the academy and the industry have shown interest in developing swiftly deployable and cost-effective delivery systems that rely on drones. This work provides a comprehensive review of the current state of Drone-based Delivery Systems. It explores the latest advancements, methodologies, and employed techniques with a specific emphasis on optimizing task scheduling and designing efficient routes for drone deliveries. It critically analyzes the advantages and drawbacks of existing solutions, contributing a thorough classification based on crucial system features, such as fleet composition, degree of device cooperation, and underlying infrastructure.

INDEX TERMS Drone delivery, drone route planning.

I. INTRODUCTION

In the latest years, the popularity of delivery services has risen exponentially.

The pandemic outbreak of COVID-19 in 2020 exacerbated this phenomenon. The increased diffusion of remote working created new necessities for the home delivery of documents or work equipment. As a result, the concept of delivery services being conveniently accessible at one's doorstep is gaining widespread acceptance among people. Nowadays, it is clear that society will not revert to previous habits, but rather we will gradually transition to a new normal state in which the popularity of delivery services will increase steadily. Currently, ground vehicles are used extensively despite their high environmental impact on traffic congestion and pollution.

In an effort to mitigate the environmental impact of logistics systems, various companies, including Amazon [69],

Google [79], and DHL [72], are actively exploring the utilization of low-emission modes of transport. These encompass bicycles [35], [36], [37], autonomous robots [38], [39], public transportation [40], [41], and Unmanned Aerial Vehicles (UAVs) or drones. Additionally, the implementation of lockers [42], [43], [44] is being considered as a means to alleviate urban traffic and its associated emissions indirectly.

Aerial vehicles represent a promising technology that is currently applied also in many other scenarios, including target surveillance [45], search and rescue operations [46], airborne communication networks [47], entertainment [48], and medical delivery [49]. Drone technology also plays a role in Industry 5.0, given that drones can be employed, for example, in the agricultural production process to improve food security [50] and environment monitoring alongside smart city infrastructure [51].

In this study, we examine existing research on Drone Delivery Systems (DDSs), focusing on the specific issue of task assignment and *trajectory planning*, i.e., assigning

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deliveries to drones and determining the optimal paths for drones, respectively.

We provide a novel classification for DDSs, considering the number of drones involved, the volume of delivery demands, and the supporting infrastructure (such as depots and support stations). We selected recent works addressing routing problems with drones applied to parcel delivery. We focus on those in which drones are expected to accomplish deliveries autonomously and without relying on additional ground vehicles, such as trucks and public transportation.

To the best of our knowledge, this is the first attempt to classify this type of system. We aim to provide useful perspectives for future research evidencing open research challenges. The main contributions of this paper are the following:

- 1) it thoroughly surveys the most recent literature addressing **route planning** and **task assignment** for drone delivery systems and identifies the challenges related to drone delivery modelling and the objectives of interest for assessing a drone delivery service;
- 2) it provides a description of system components and proposes a novel classification based on such infrastructures;
- 3) it surveys solutions adopted by the industry, highlighting the growing interest in this field;
- 4) it provides a brief summary of different topics of interest other than drone route planning which are studied in the literature, including modern delivery systems that use other technologies in addition to drones.

This survey is organized as follows. Section II provides a brief overview of surveys conducted on related topics. In Section III, we discuss some statistics about the reviewed articles. Then in Section IV, we present our classification of DDS and position recent contributions in the literature. Moreover, in Section V we outline the challenges common to all DDSs. In Section VI, we draw our conclusions. In Appendix A, we discuss some trials and applications available in the logistics industry. In Appendix B, we provide a brief summary of other delivery systems that are of growing interest to the research community, and finally, in Appendix C, we report additional challenges that come with the design of drone delivery systems and which are not strictly related to the route planning problem.

II. RELATED WORK

Parcel delivery systems raised attention in the research and industry community in the past years. Other researchers have surveyed existing solutions proposed in the literature. Some surveys study articles focusing only on hybrid-truck drone delivery systems [52], [53]. In contrast, in this paper, we review papers that propose solutions for drone-only autonomous delivery.

Boysen et al. [54] present a survey on last-mile delivery systems from an operations research perspective, considering

different means of transportation. Most of the articles considering drones propose a multi-modal solution, i.e., a combination of drones and trucks to perform deliveries. The authors do not focus on trajectory planning, but they consider a wide variety of problems, including but not limited to trajectory planning.

Chung et al. [55] survey optimization approaches aimed at solving drone and drone-truck combined operations covering several application fields such as transportation and logistics, construction and infrastructure, security and disaster management, agriculture, etc. The work addresses several problems, such as trajectory planning, area coverage, searching operations, scheduling, task assignment, communication protocols, data gathering, etc. Few of the reviewed articles address the drone delivery problem from a task assignment and trajectory planning perspective.

Benarbia et al. [56] investigate many challenges related to drone delivery, such as drone routing, task assignment, recharging station deployment, landing issues, and fleet sizing. In addition, a short review of drone applications in the logistics industry is also provided. Moreover, the authors discuss the feasibility of implementing drone delivery services in urban areas.

Similarly to our paper, the survey proposed by Moshref-Javadi et al. [57] focuses on route planning in drone-based delivery systems. The authors reviewed both works in which drones perform deliveries autonomously and works in which drones perform deliveries relying on other vehicles. Also, a multi-criteria classification of models has been proposed. Firstly, according to their classification, a model can be defined as i) “Pure-play Drone-based”, ii) “Unsynchronized Multi-modal” iii) “Synchronized Multi-modal”, and iv) “Resupply Multi-modal”. Categories ii), iii), and iv) consider models with other vehicles that play a supporting role for drones; thus, they are beyond the scope of our survey. Category i) regards models in which drones perform deliveries autonomously but rely only on some of the types of infrastructure we consider; thus, it partially covers the models of our interest. Our survey differs from the one proposed in [57] first because we have selected more recent works, and second because some of the works we have selected envisage systems that do not comply with the classification proposed by Moshref-Javadi et al., such as the ones relying on support stations.

Pasha et al. [58] review drone scheduling problems, including route planning. The authors categorized 145 reviewed studies into different categories such as i) general drone scheduling, ii) drone scheduling for delivery of goods, and iii) drone scheduling for monitoring. For each category, the authors provided a useful representative mathematical model through MILP formulations. The model proposed for drone delivery is not representative of all the scenarios addressed in the literature and our study. When surveying the context of delivery systems, the authors of [58] consider a wide variety of works, including multi-modal solutions in which drones rely on ground vehicles.

TABLE 1. Related work summary.

Reference	Focus	Findings
Boysen et al. [54]	Multi-modal last-mile delivery	Summary of currently applied delivery modes (manned vans, bikes); near future modes (technology is available but applied only in field tests, such as drones, autonomous delivery robots, crowd-shipping); farther future modes (missing technology, such as autonomous driving, and tunnel-based cargo).
Chung et al. [55]	Drones and drones combined with trucks applied to several scenarios (not only parcel delivery).	Identifies barriers to drone and drone-truck operations' implementation (such as, privacy, security, safety, environmental concerns and socio-economical factors).
Benarbia et al. [56]	Multiple challenges related to drone delivery.	Identifies drones as ideal for last-mile urban delivery pointing out some limitations to overcome (such as, fleet sizing, scaling issues, limited real-data-based simulations).
Moshref-Javadi et al. [57]	Route planning for drone delivery with/without support vehicles.	Identifies the lack of comprehensive real-world case studies, realistic drone parameters and urban/suburban/rural settings. Identifies the need of addressing safety and privacy issues, and the adaption to current logistic systems.
Pasha et al. [58]	Drone scheduling in several application scenario.	Drone scheduling limitations should be addressed through consultation with industry and experts. Lack of works addressing issues such as collaborative scheduling, continuity of service and adaptation to dynamic environments.
Macrina et al. [59]	Operational research to solve route planning for drone delivery with/without support vehicles.	Identifies lack of research on energy consumption, CO2 emissions, uncertain data and dynamic systems.

Macrina et al. [59] focus on operational research techniques to solve routing problems for parcel delivery. The authors provide a structured literature review in which 63 articles published between 2015 and 2020 have been categorized based on which types of equipment are used to accomplish deliveries. Specifically, the authors define two macro-categories: the first in which deliveries are performed by both trucks and drones and the second in which they are performed by drones only.

Focusing on route planning of drones in delivery systems, this survey is the closest to our work. However, differently from [59], we focus on delivery systems exploiting only autonomous drones as means of transportation. Moreover, we consider a wider variety of methodologies. Furthermore, our taxonomy generalizes the one proposed by [59].

In Table 1 we summarize the focus and findings of related work.

III. SELECTED ARTICLES

To conduct this survey, we used Google Scholar, Scopus, IEEE Xplore, ACM Digital Library, and the search tools provided by Litmaps [60]. We searched for the following keywords: Drone Delivery, Drone Delivery System, Route Planning, and Trajectory Management.

Since we focus on route planning in drone delivery systems, we excluded all the articles which tackle different aspects of drone delivery or consider additional vehicles other than drones.

To capture the growing interest in drone-based delivery systems, we show some statistics provided by Scopus. We considered articles published from 2014.

Figure 1 shows the number of published articles that we found performing the query “Drone AND Delivery AND (Route AND Planning) OR Routing”. Looking at search results, it is possible to observe that the trend is exponentially growing over time; thus, confirming the growing academic interest in drone technology applied to delivery services.

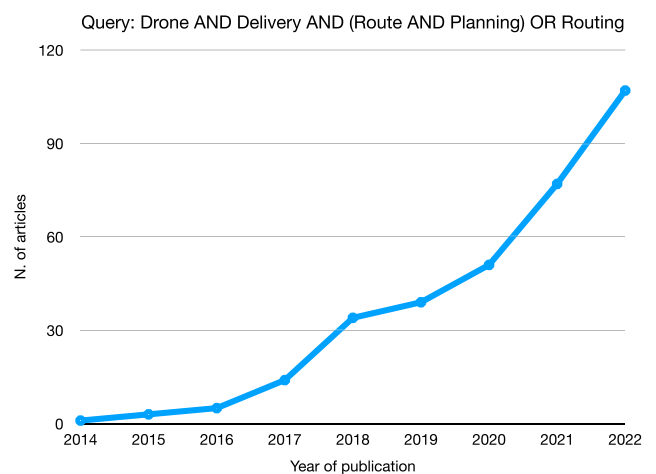


FIGURE 1. Keyword analysis: results obtained searching “Drone AND Delivery AND (Route AND Planning) OR Routing” on Scopus. Number of articles vs. publication year.

Removing the articles concerning multi-modal solutions or those not focusing on trajectory planning and scheduling, we selected 34 papers published between the years 2017 and 2022. The chart in Figure 2 shows the number of selected articles per year of publication.

In Table 2, we summarize all the acronyms used throughout the paper.

IV. CLASSIFICATION

We classify the literature on route planning for delivery systems by considering four classes of approaches, differing in terms of the volume of delivery demand and the number of drones: 1) Single Drone, which is subdivided into Single Delivery and Multiple Deliveries; and 2) Multiple Drones Multiple Deliveries. Each class can be further divided into three sub-categories depending on which infrastructure the delivery services rely on, namely Single Depot, Multiple

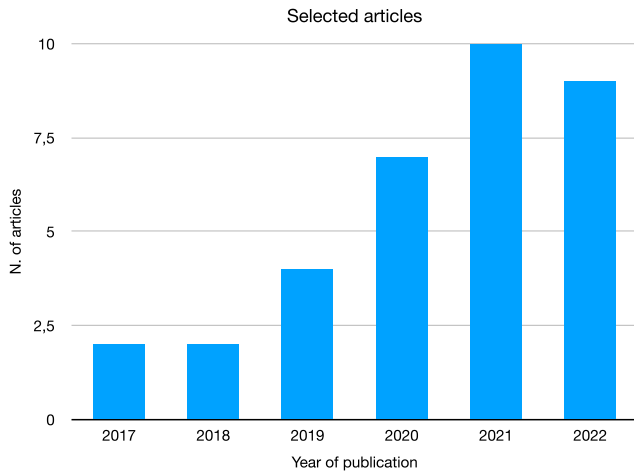


FIGURE 2. Selected articles: number of articles vs. publication year.

Depot, and Support Stations. A schematic representation of our classification is depicted in Figure 3.

A. SYSTEM COMPONENTS

Before delving into the different classes, we present the system components and define some terminology:

- **Parcel:** package that must be delivered. It may have a restricted size and weight due to technical specifications or design choices.
- **Drone:** transportation mean used to accomplish deliveries. DDSs inherit the limitations of drones; e.g., payload capacity, flight speed, flight range, battery management, etc. If the DDS relies on a fleet of drones then coordination among drones has to be considered and handled.
- **Support Station:** decentralized entity deployed in the area of interest and designed to provide assistance to drones during their missions. Several support stations might be deployed to establish a transportation network. A support station might be conceived to be a designated place where shippers can drop off the parcels to be delivered, and/or receivers can pick up their parcels. Moreover, to cope with drones' energy limitations and allow long-distance deliveries, support stations may be equipped with some mechanism that allows drones to recharge their battery or replace their low battery with a full one.
- **Depot:** central entity designated to assist drones in their operations, e.g., allowing battery recharge/replacement and parcel on-board loading. Moreover, contrary to support stations, parcels can be stored and receivers cannot directly pick up a parcel at the depot. Also, the depot is the starting and ending point of drone routes, but not an intermediate pit-stop as in the case of support stations.

TABLE 2. Acronym table.

	Acronym	Description
<i>Architectural features</i>	S	Single delivery
	M	Multiple delivery
	SD	Single Depot
	MD	Multiple Depots
	SS	Support Stations
	SP	Suppliers
<i>Objectives</i>	T	Time
	C	Cost
	RU	Resource Utilization
	D	Distance
	DP	Delivered Parcels
	L	Loss due to drone failure
	OT	On Time delivery
	SR	System Reliability
<i>Energy limitations</i>	MFT	Maximum Flight Time
	MFD	Maximum Flight Distance
	BC	Battery Capacity
<i>Energy limitations</i>	CM[...]	Consumption model with ...
	PW	Payload Weight
	DW	Drone Weight
	BW	Battery Weight
	DR	Drag force
	W	Wind
	+	other
<i>Actions</i>	REP	Replace
	REC	Recharge
<i>Uncertainties</i>	W	Wind
	DF	Drone Failure
<i>Methodology</i>	MILP	Mixed Integer Linear Programming
	MIP	Mixed Integer Programming
	CSP	Constraint Satisfaction Problem
	H	Heuristic
	GPH	Graph Algorithm
	DP	Dynamic Programming
	GR	Greedy
	LS	Local Search
	GEN	Genetic algorithm
	BKC	Blockchain
	DaaS	Drone as a Service
<i>Data source</i>	SYD	Synthetic Data
	RD	Real Data

- **Supplier:** storage facility where parcels are stocked and withdrawn by drones. It does not provide drone assistance.
- **Customer:** person/entity who is meant to send or receive the parcel. When initiating the delivery request, the location of the parcel's destination must be provided. Such location can, according to the system design, either be an arbitrary location (e.g., their house, a residence building, a high-rise building, a remote cabin, etc.) or a predefined location (e.g., a support station).

With reference to the legend of Figure 4a, Figures 4b, 4c, 4d, and 4e illustrate the infrastructure of possible scenarios in which multiple drones handle multiple deliveries.

In most of the systems proposed in the literature, drones rely on a centralized infrastructure. This is the case of the scenarios shown in Figure 4b, 4c and in Figure 4d. In Figure 4b there is a single depot, where parcels are stocked and then loaded onto drones when ready to be shipped.

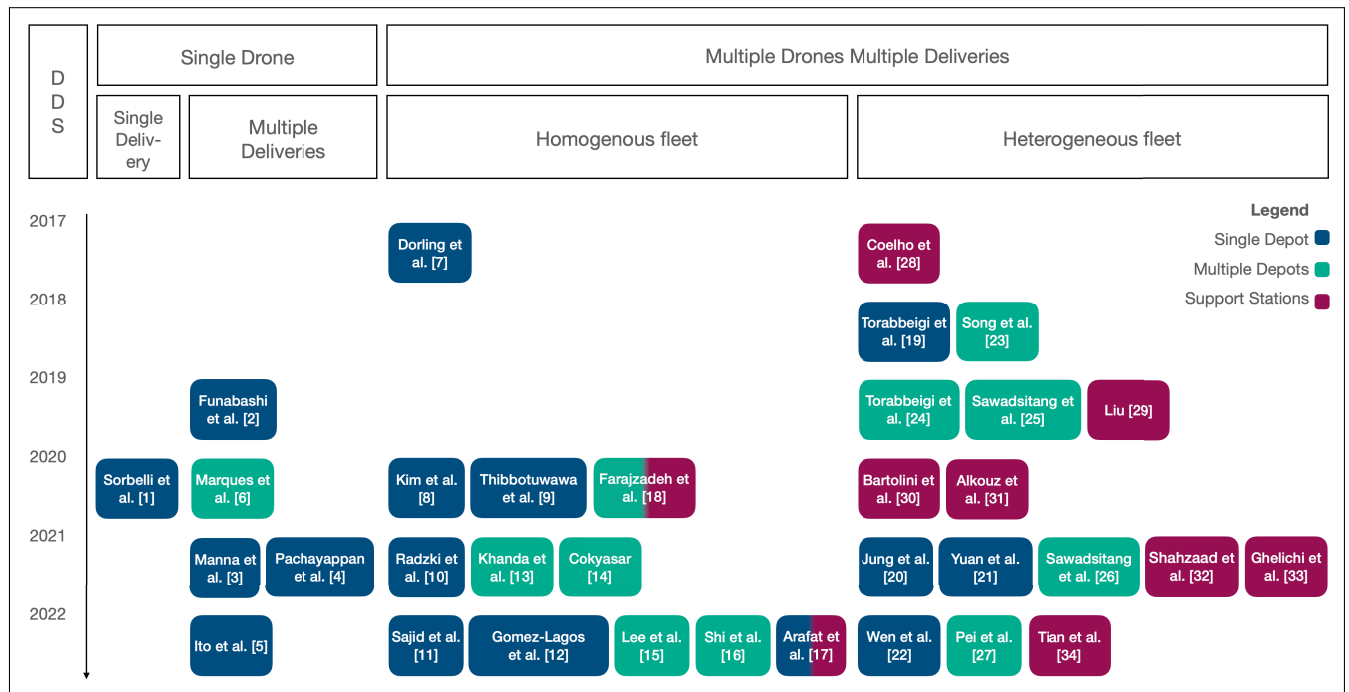


FIGURE 3. DDS classification.

Moreover, drone batteries can be replaced or recharged at the depot. All drones move back and forth from the central depot to arbitrary locations provided by customers. Similarly, Figure 4c represents a variation of the single depot scenario that includes suppliers. The drones' batteries are still charged or replaced at the depot, but parcels to deliver can also be stored in supplier facilities. In Figure 4d, there are multiple depots and multiple drones, but each drone is assigned to a *base depot*. Thus, drones move back and forth from their own base depot to customer locations. Also, in this case, drone batteries can be replaced or recharged at depots. In some cases, the possibility of transferring parcels between depots or outsourcing deliveries to external carriers is also envisaged. A few articles propose a decentralized infrastructure, as depicted in Figure 4e. In this scenario, support stations are deployed in the service area, making up a transportation network that drones exploit to replace or recharge their batteries. Moreover, some authors also formalize drones' cooperation for handing over parcels at support stations.

B. SINGLE DRONE

In this section, we describe drone-delivery solutions considering a single drone. Table 3 reports salient features of such works.

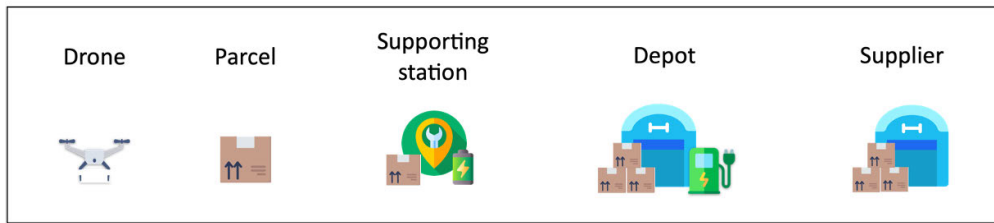
1) SINGLE DELIVERY

The problem addressed in the single drone delivery scenario consists of planning a single route to carry out a single delivery request. To our knowledge, this scenario has been

considered only in a single depot infrastructure. Thus, finding a solution to this problem, from a routing point of view, is trivial and consists of computing the trajectory from the depot to the customer's location and then back to the depot. If no other issues are addressed (e.g., uncertainties due to environmental conditions or drone failures), this problem comes down to the ability of the drone to fly to a given location.

Sorbelli et al. [1] address Single Drone Single Delivery considering the effect of wind on energy consumption. Wind conditions are monitored by wind control units deployed in the region to provide updated measurements when the drone approaches them. The area of interest is modelled as a time-dependent cost graph whose vertices are the drone's depot and the wind control units. The cost of an edge represents the energy consumed by the drone to traverse it and depends on the speed and direction of the wind. Given a customer order, the mission budget, and the initial edge costs, the addressed problem consists of finding the least-cost cycle between the depot and the customer location without stopping at intermediate vertices, allowing the drone to change its trajectory depending on weather conditions. In order for the solution to be successful, it should provide a route that fits the given budget. The authors propose three algorithms. The Offline Shortest Path (OSP) algorithm finds the best cycle with initial conditions checking that its cost is below the budget. Upon cost updates, two online algorithms are proposed to recompute the best solution: the Dynamic Shortest Path (DSP) and the Greedy Shortest Path (GSP). The former finds the new least-cost cycle whereas

Legend



(a) Legend of system components' icons.



(b) DDS handling multiple deliveries with multiple drones that rely on an infrastructure made of a single depot.



(c) DDS handling multiple deliveries with multiple drones that rely on an infrastructure made of a single depot and some suppliers.



(d) DDS handling multiple deliveries with multiple drones that rely on an infrastructure made of multiple depots.



(e) DDS handling multiple deliveries with multiple drones that rely on an infrastructure made of support stations.

FIGURE 4. Working scenarios of DDS.

the latter recomputes the new route on the basis of the next vertex with the least cost. The authors consider the possibility that the drone might not be able to fly back to the depot, nor to accomplish the delivery or take off from the depot at all due to unexpected wind conditions. The authors conducted experiments both on real and synthetic data. All the proposed algorithms (OSP, DSP, and GSP) have been tested and compared. The results show that GSP is the least performing among the proposed algorithms.

DSP completes more deliveries than all the approaches but with more failures than the OSP, which is the most conservative approach as it might decide to cancel some deliveries. The major limitation of this work is that battery consumption and capacity constraints are not considered, which is a very unrealistic hypothesis. Moreover, the work does not propose any strategy to tackle the return to the depot which could be necessary in case of unexpected battery depletion due to unforeseen adverse weather conditions.

2) MULTIPLE DELIVERIES

In the case of a single drone carrying out multiple deliveries, the planned route has to fulfil multiple delivery requests. The drone might be loaded with multiple parcels or one at once. The drone route can be a single trip in the former case and a multi-trip in the latter. The majority of works that address problems falling in this category propose systems relying on a single-depot infrastructure. However, there is one work that considers support stations.

a: SINGLE DEPOT

Funabashi et al. [2] formulate the route planning problem as a MILP that extends the TSP formulation. The authors named the problem FSVRP. The goal is to find a route starting from the depot, reaching all customers, and returning to the depot in the minimum time. As a drone can carry multiple parcels, it is important to consider how the weight of parcels affects drone performance. This paper considers the impact of the payload of a drone on its flight speed. Specifically, given the drone driving force and the velocity with no payload, the authors derive an equation that expresses speed as a function of the payload. It is assumed that the parcels' total weight is below the drone's carrying capacity limit. Energy consumption and battery limitations are not considered. To solve the optimization problem, the authors propose a Dynamic Programming algorithm. Experiments are conducted on synthetic data. The authors considered several scenarios with a variable number of deliveries. In order to compare the proposed DP algorithm, several benchmarks are considered: a brute force algorithm to optimally solve FSVRP and a brute force algorithm to optimally solve the TSP problem, which both could not be solved within an hour instance with more than 12 customers; a simple heuristic algorithm and the nearest-neighbor algorithm for TSP, which have similar performance and both are outperformed by every other tested algorithm; and finally a DP algorithm for solving TSP, which performed worse than the proposed DP algorithm for FSVRP. In particular, for small scenarios (up to 11 deliveries), the work also considers a brute-force algorithm to find the optimal solution. Although tests comparing several algorithms have been carried out, the considered instances are too few and too small for a comprehensive assessment of the proposed approach.

Manna et al. [3] address the problem of minimizing the total distance covered by the drone so as to meet all customers' demands. The authors propose two models: i) NAPD (Not Allowing Partial Delivery), in which the demand of customers is always meeting the drone payload capacity constraint, thus customers need to be visited only once; ii) ADP (Allowing Partial Delivery), in which the demand of customers may exceed the drone payload capacity. Thus it may be necessary to split the demand into multiple orders and visit the customers multiple times. Each model is solved with a greedy approach. The greedy algorithm solving the NAPD model computes multiple trips for the

drone starting and ending at the depot and reaching a variable number of customers. Each trip is built in a greedy approach, by considering the closest customer to the depot at the first iteration and the closest to the previous customer in the trip for the following iterations. Customers are included in a trip until the drone payload capacity is not exceeded. The greedy algorithm solving the APD model works in the same manner. Still, it considers separately the items making up the demand of each customer, and when including a customer in a trip, its non-delivered items are selected in payload ascending order up to the drone payload capacity. Authors conducted experiments considering randomly generated sets of instances with an increasing number of demands. They compared their solution with a baseline approach in which the drone can serve a single customer per trip.

Pachayappan et al. [4] study the Drone Routing Problem with Pickup and Delivery (DRPPD) under the assumption that a single drone has to carry out multiple deliveries in the presence of a single depot. The depot can be used only for recharging purposes whereas multiple suppliers provide storage, pick-up and drop-off services. The authors formulate a MILP to minimize the total distance travelled by the drones. The idea is the following: a single drone can be charged at the depot, where services are assigned to the drone. The drone can transport only one parcel at a time. During the first service, the drone travels from the depot with a fully charged battery to the pick-up point of the first delivery and from there to the delivery point. Before accepting to serve a delivery, the drone must compute how much energy such travel will consume and whether it has enough power after the shipment is completed for a safe return to the depot. If the remaining battery level is enough to accomplish the next delivery and return safely to the depot, the drone will travel to the next pick-up point. Otherwise, it will return to the depot. The problem is then solved with a Drone Neighbor Search Heuristic (DNSH), which schedules the next delivery as the one requiring minimum distance. The experimental results show the relative percentage gap in the performance obtained by solving the MILP and implementing the DNSH. Data is randomly generated and the tests show that the proposed heuristic has a relative gap from 1.89% to 50.62%. One limitation of this work is that the authors performed tests considering unrealistic scenarios. For example, they consider an area of more than 1000 km in diameter to be covered with a single drone, meaning that even if the drone could fly uninterruptedly at an average of 14 km/h, it would take more than 3 days to accomplish the scheduled deliveries.

Ito et al. [5] extend the formalization they had introduced in [2] by considering both the weight of the load and wind conditions while minimizing the flight time. They call this problem the Flight Speed-aware Vehicle Routing Problem with Load and Wind (FSVRPLW). A drone departs from a single depot with N parcels to be delivered to N customers. The drone can be charged only once, at the departure from the depot. In FSVRPLW, drones' velocity is modelled as a function of the parcel's weight

TABLE 3. Salient features for articles addressing delivery with a single drone.

Reference	Deliveries	Ground components	Objectives	Energy limitations	Consumption model	Battery Management	Uncertainties	Methodology	Simulation data
Sorbelli et al. (2020) [1]	S	SD	C	BC	CM[PW, DW, DR]		W	GPH	SYS, RD
Funabashi et al. (2019) [2]	M	SD	T,D					DP, MILP	SYD
Manna et al. (2021) [3]	M	SD	D					H	SYD
Pachayappan et al. (2021) [4]	M	SD	D	BC	CM	REC		H, MILP	SYD
Ito et al. (2022) [5]	M	SD	T	MFD	CM[P, W]		W	DP	SYD
Marques et al. (2020) [6]	M	SS	T, RU	SD	CM	REC		GR, LS	SYD

(which decreases after each delivery) and of the wind. Similarly to the approach implemented by the same authors in their previous work [2], the FSVRPLW problem is solved with a dynamic programming algorithm. Each state is characterized by the set of already visited customers S and the index of the last visited customer i . The transition from a state (S, i) to another state is given by the minimum time required to travel from customer i to customer $i' \notin S$. This dynamic programming algorithm is compared to other algorithms that solve TSP, FSVRP [2] and FSVRPLW without parcel weight modelling. The results show that the normalized flight time and distance do not improve significantly compared to the benchmark algorithms.

b: SUPPORT STATIONS

For the single drone scenario, the literature does not consider multiple depots, but it does consider support stations. This allows drones to reach long distances when considering energy limitations. Marques et al. [6] introduce a multi-objective problem that aims at minimizing energy consumption and delivery time. In doing so, two constraints must be satisfied: a drone should reach a support station to recharge its battery/fuel if its residual energy is not enough to reach the next customer and to come back safely to a support station, and drones should not fly above no-fly zones in urban areas. To tackle this problem, the authors implement a metaheuristic that combines together a Greedy Randomized Adaptive Search Procedure (GRASP) and Multi-Objective Variable Neighborhood Descent (MOVND) algorithms. The GRASP algorithm generates an initial solution that is then passed to MOVND. Given the drone's current position, GRASP iteratively chooses the k closest clients and randomly selects one of them as the next node to visit. MOVND works by applying a local search method where, given the current solution x , a set of neighbour operations are defined. Some are applied to x to obtain x' , which is discarded if it does not improve x , and it is used as the next best solution otherwise. The authors evaluate the proposed methods in different settings, i.e., by changing parameters and the order of the operations, by means of hypervolume and percentage and the absolute value of Pareto-optimal solutions. The authors do not

provide a study of the related work and do not highlight how their model advances the state-of-the-art. The performance evaluation of the proposed algorithm is not compared with any other existing solutions.

C. MULTIPLE DRONES MULTIPLE DELIVERIES

The multiple drones multiple deliveries is the most complex scenario and also the most addressed in the literature.

Typically, each delivery is assigned to a single drone, which has to carry the parcel from the source to the destination. Cooperation among drones, allowing parcels to be exchanged between drones, has been proposed only in two articles [28], [30].

The fleet of drones may be made of identical drones (i.e., homogeneous fleet), or made of drones with different specifications such as payload capacity, battery capacity, size etc. (i.e., heterogeneous fleet). We group together works depending on whether they consider homogeneous or heterogeneous fleets.

For this scenario, the literature proposes different setups, which include: i) single depot, ii) multiple depots, iii) presence of support stations, and iv) mixed. In the following sections, we provide several tables summarizing salient features of the reviewed works.

1) HOMOGENEOUS FLEET

Table 4 summarises salient features for works addressing Multiple Drones and Multiple deliveries with a homogeneous fleet of drones. For the acronyms we refer the reader to Table 2.

a: SINGLE DEPOT

Dorling et al. [7] address the problem of minimizing either the cost in terms of energy consumption and fleet size or the time required to deliver multiple parcels to different customers, planning both the number of drones to include in the homogeneous fleet and their routes. Each route starts and ends at the depot that is used both to replace batteries and pick up packages. The number of drones is minimized by reusing the same drone for multiple routes when possible. The authors include models for drones' energy consumption,

TABLE 4. Salient features for articles addressing delivery with multiple drones with homogeneous fleet.

Reference	Ground components	Objectives	Energy limitations	Consumption model	Battery Management	Uncertainties	Methodology	Simulation data
Dorling et al. (2017) [7]	SD	T, C		CM[PW,DW,BW]	REP		MILP, H	SYD
Kim et al. (2020) [8]	SD	nDP	BC, MFT	CM	REC		MILP, H	RD
Thibbotuwawa et al. (2020) [9]	SD	nDP	BC	CM[PW, DW, DR]		W	CSP	SYD
Radzki et al. (2021) [10]	SD	nDP		CM[PW, DW, DR]		W	CSP	SYD
Sajid et al. (2022) [11]	SD	T	BC	CM[PW]	REP, REC		MILP, H	SYD
Gomez-Lagos et al. (2022) [12]	SD	D					CSP, H	SYD
Khanda et al. (2021) [13]	MD	C		CM[PW, W]	REP, REC	W	GPH	SYD
Cokyasar (2021) [14]	MD	T	MFD		REP		MILP	SYD
Lee et al. (2022) [15]	MD	RU	BC	CM[PW, W]			H	SYD, RD
Shi et al. (2022) [16]	MD	T, C, RU	MFT, BC	CM[PW, BW]			MILP, GEN	SYD
Arafat et al. (2022) [17]	SD, SS	T	MFD	CM[PW, W, DR]	REC	W	MILP, H, GPH	SYD
Farajzadeh et al. (2020) [18]	MD, SS	C	BC	CM	REC		MILP	SYD

battery weight, which is proportional to its residual charge, and payload weight. The problem is first formalized as a MILP, and then a simulated annealing heuristic algorithm is proposed to find suboptimal solutions simulated in several scenarios. Firstly, the work proposes an energy consumption model based on real measurements taken with and without payload. Then, the authors show that results obtained with the Simulated Annealing algorithm consistently find near-optimal solutions for small scenarios, which include 6 to 8 delivery requests. The Simulated Annealing algorithm has also been used to solve larger instances including 125 and 500 delivery requests. This used to be considered a pioneering work in the context of drone delivery and still remains relevant and is used as a state-of-the-art comparison for more constrained drone delivery problems.

Kim et al. [8] consider a fleet of homogeneous drones to deliver parcels from a central depot to customers' locations. Customers may request multiple deliveries, but drones can pick up only one package at a time at the central depot. The objective is to find a multi-trip route for each drone, maximizing the number of delivered parcels within a given temporal horizon. The energy consumption is modeled without considering the effect of the parcel weight. To prevent battery depletion, the work adopts a conservative approach and considers a worst-case battery consumption ratio (kWh) to estimate the energy consumed to perform deliveries. In addition, the work assumes fully loaded drone batteries, which could hamper the feasibility assessment of delivery tasks. However, to address battery limitations, the approach considers recharging stations at the depot. The problem is formalized as a MILP and is solved using a heuristic algorithm. To validate the proposed approach, experiments with realistic settings are presented. The authors selected a service area that corresponds to a district in Seoul. Delivery destinations correspond to rooftops of buildings in the service area, and the central depot is located in the post office.

Several scenarios have been considered: small scenarios, with two drones and ten delivery destinations; medium scenarios, with four drones and 20 destinations; large scenarios, with ten drones and 50 destinations. Each delivery location has a demand ranging from 1 to 5 items. For each scenario, 20 random instances have been generated. Heuristic solutions have been compared to optimal solutions found by the Gurobi solver. Gurobi solver managed to solve in a reasonable amount of time all the instances of small and medium scenarios, whereas, for large scenarios, only 2 instances were optimally solved within 12 hours. Both for small and medium instances, heuristic algorithms have found solutions as good as the optimal, but the authors do not provide optimality proof.

Thibbotuwawa et al. [9] propose a two-level planning approach for UAV mission planning. At the *Mission Planning* level, the transportation network is divided into a set of clusters, each including the base depot and has a homogeneous fixed-size fleet of drones. The *Sub-Mission Planning* level concerns the definition of drone routes. The objective is to meet the customers' satisfaction level requirement by finding routes starting and ending at the base depot. The overall solution combines *Sub-Mission Planning* solutions. The work proposes two different strategies depending on whether the airspeed or the ground speed of the fleet of drones is constrained to be constant. In the first strategy, the ground speed is assumed to be constant, and thus the airspeed needs to be adjusted to compensate for the wind effect. The second strategy assumes that the airspeed is constant; thus, the ground speed is calculated according to wind speed, and what needs to be adjusted is the heading angle so that the desired location is reached. The authors show that the strategy that keeps the airspeed constant is more efficient in terms of power consumption, but it usually takes more time to complete a multi-trip to the customers' delivery destinations. In contrast, the strategy that keeps a constant ground speed is usually less

efficient in terms of power consumption, but it achieves to complete the deliveries earlier. An extension of this work is presented in Radzki et al. [10], where several applications of the proposed model are presented considering a disaster relief scenario. Among the applications proposed, there is the feasibility assessment of existing plans under given wind conditions; the identification of non-accessible areas that, without taking into account winds, would have been reachable by drones; the planning of emergency routes to return to the depot in case the actual wind conditions are worst than the forecast. The authors in [9] consider scenarios with different wind speeds and angles (10 m/s 30°, 11 m/s 130°, 12 m/s 230°), fleet size (2, 3, 4) and delivery demand (4, 6, 8, 10). The objective is to compare the two proposed strategies for computing energy consumption considering wind effects. The results show the energy consumed given the scenario and under varying wind conditions. The authors in [10] consider a scenario with 10 deliveries and 4 drones to show some of the proposed applications of the model. Given a feasible solution to the problem, the authors show how to find plans that are robust to the given wind conditions, how to identify inaccessible areas due to wind conditions, and how to plan emergency returns of UAVs in case real wind conditions are worse than the forecast. The authors provided more extensive simulations to demonstrate the applicability of the model to online decision-making. For these experiments, the authors considered several fleet sizes (2, 3, 4), demand volumes (4-12), and wind speeds and directions (9, 10, 11 m/s [0°, 360°)).

More recently, Sajid et al. [11] formalize a joint optimization of UAVs' routing and route scheduling problems. The UAV-routing problem involves finding the best routes for geographically distributed customers to deliver their demands while minimizing travel time. The UAV-route scheduling problem aims at minimizing the makespan (i.e., the overall time to complete all the demand deliveries) by scheduling aerial routes with a fixed number of drones, m . Firstly, the authors model the UAV-routing problem, which minimises the sum of the time for completing the deliveries assigned to each drone while meeting the payload and battery capacities, flow, and integrity constraints. The drones depart from a single depot to deliver possibly more than one demand each. The depot is the only charging station in the network. The authors introduce the notion of *aerial route*, also referred to as batch-of-routes (BoR), a geographical partition of the set of costumers, starting and ending in the depot. When formalizing the UAV-route scheduling problem, the authors notice that due to the limited number of drones available at a depot, a single drone might be necessary to execute multiple aerial routes. The UAV-route scheduling problem involves minimizing the maximum expected completion time of the aerial routes of each drone, which is defined as the sum of the expected ready time of the assigned drone and the expected time required to execute the aerial route. In particular, the expected ready time of a drone is the time instant when all previously scheduled aerial

routes have been executed, and the drone is ready to execute the next aerial route, i.e., it is fully charged at the depot. The authors propose two algorithms for solving the two described problems. A hybrid genetic and simulated annealing approach (HGSA) is proposed for the UAV-routing problem. The UAV-route scheduling problem is solved by a UAV-Oriented MinMin (UO-MinMin) algorithm, which assigns delivery missions to available drones as soon as they become available. Experiments are simulated on a dataset of 76 nodes, one depot and 75 customers. HGSA algorithm sensitivity is studied through Monte Carlo and then compared to other benchmarks, namely Particle Swarm Optimization & Simulated Annealing algorithm (PSO-SA), Differential Evolution & Simulated Annealing (DESA), and Harris Hawks optimization (HHO) algorithms, showing improved performance in terms of power consumption and travel time. The OU-MinMin algorithm performance is compared to two heuristics designed to solve the independent task assignment problem; namely, the minimum completion time (MCT) and opportunistic load balancing (OLB) algorithms [61], [62], resulting in a smaller makespan in all considered configurations.

Gómez-Lagos et al. [12] formalize three versions of the Pickup to Delivery Drone Routing Problem (PDDRP), whose objective is to find an assignment of drones to customers such that all the customers are served while minimizing the makespan, which in this case is defined as the maximum distance travelled by a drone. The first formulation models the problem as a m traveller salesman problem (m -TSP). The goal is to find one route with minimal cost for each one of m salesmen (i.e., drones), who start and end their delivery at the depot, so each customer is visited exactly once. The second formulation is a variant of the Parallel Machines Scheduling problem; each job can be represented as a network node. The completion time of jobs translates into the distance travelled on serving the customers and landing at the depot. Finally, the third formulation involves a single depot, a set of facilities (i.e., suppliers), a set of customers, a homogeneous fleet of drones, and customer orders. The scheduling must be optimised to minimize the makespan, considering that each customer has only one demand. The authors prove that PDDRP is NP-hard and propose variants of the GRASP metaheuristic for finding solutions efficiently. GRASP iteratively calls a greedy randomized strategy for finding a feasible solution and a local search procedure to improve it. The experiments are conducted on synthetic data by varying the number of customers, suppliers and drones. The authors use CPLEX solver [63] to find the exact solutions of their MILPs for small instances and then evaluate the result of the GRASP variants. The experiments highlight that no one formulation outperforms the others in all considered scenarios, even though the third formulation performs almost optimal solutions (gap < 0.1%) for small-sized instances. This is because the third formulation also accounts for supplier facilities. One limitation of this study is that it offers two formulations referring to well-known

scheduling problems for which approximation bounds have been proved, but does not explain how these results relate to their own problem.

b: MULTIPLE DEPOTS

Khanda et al. [13] address the problem of minimizing the energy consumption of a fleet of homogeneous drones which operates under uncertain wind conditions. The authors modelled the problem considering dynamic time-dependent graphs, using a similar approach to Sorbelli et al. [1]; this work considers the displacement of wind control units to gather wind measurements to update edge costs over time. However, multiple depots are considered to provide drone assistance (e.g., battery replacement/recharge) and parcel storage. Each depot has its own fleet of drones and serves a set of customers. To model this scenario, the authors define a dynamic graph as follows: nodes represent the depots, the wind control units and the customers; edges are asymmetric and represent the possible drone flights; edges' weight is the energy required by the drone to move from one point to another. The graph has a static topology, but the edges and their cost can change over time (edges can be added or removed, and their cost can be updated to reflect the wind's current conditions). Drones can serve one customer at a time. Hence, a drone route starts from its corresponding depot, reaches a customer to perform the delivery, and ends at the same depot. When computing the shortest path that the drone has to follow to complete its tour, the authors include the effect of the payload on drones' energy consumption. To cope with these constraints, the authors designed a pre-processing algorithm that computes the assignment of deliveries to drones and the initial route considering the wind conditions at the time the delivery is assigned. Customers are served in a First-Come-First-Served fashion, thus deliveries are assigned as soon as there are drones ready to depart from the depot. Given the initial assignment and the initial routes, it is necessary to consider that some routes may become unavailable or more costly due to the changes in the wind's conditions. Thus, upon wind changes, the costs of the graph edges are updated, and the routes should be changed accordingly. To address this issue, the authors propose an online centralized parallel algorithm able to recompute drone routes efficiently. The routes are recomputed by considering both the edge changes and the current positions of the drones. The authors also presented a complexity analysis showing that the time complexity depends on the number of edges added or removed in the graph and the degree and the diameter of the graph; the space complexity depends both on the number of drones and the size of the graph. Moreover, the authors computed the speedup of their parallel approach, which they showed to be at least the graph's diameter multiplied by the number of processing units utilized. Finally, the authors evaluated their approach by performing tests on publicly available large graph instances [64] and comparing their results with a

parallel GPU-based algorithm provided by the Gunrock [65] library, which computes from scratch the new paths after every change in edge costs. In their tests, edges are randomly added and removed as edge change batches, and the execution time to perform the update is recorded. Three scenarios are evaluated: in this first scenario, the batch has more edge insertions; in the second one, the batch has more deletions; finally, the authors consider the scenario where insertions and deletions are perfectly balanced. Results show that the execution time of the proposed approach is suitable for an online algorithm, being approximately 100ms on average. Moreover, it outperforms Gunrock library, proving to be twice as fast. However, the authors have not evaluated either the reduction of energy consumption or the impact on the number of successful deliveries that their approach should bring in dealing with changing weather conditions.

Cokyasar [14] formalizes the delivery drone route planning (DDRP) problem under the assumption that several automated battery swapping machines (ABSMs) are available for drones to replace their battery throughout their delivery missions. DDRP is formalized as an MINLP (Mixed Integer Non-Linear Program), where the objective function to be minimized is a combination of the total travel time for all deliveries and the average order queueing time (including the battery replacement service time) over a planning horizon, W . The demands are assumed to arrive following a Poisson process. The first proposed solution consists of linearizing the objective function and introducing new variables that transform the original MINLP into a quadratic problem (MIQCP). Such MIQCP is solved using Gurobi. The second proposed solution exploits the convexity of W to explore refined solutions iteratively until a desired precision of the optimal solution is achieved. These solutions are evaluated in different settings by varying the number of ABSMs and demands. The solution of the MIQCP obtained by Gurobi achieves a gap to the optimal solution that spans from 10^{-15} to 6.2×10^{-2} , whereas the gap obtained by the iterative solution lies between 9.9×10^{-5} to 2.6×10^{-2} , and is up to 3.5 times faster to process.

Lee et al. [15] tackle the problem of deciding routes in hub-to-home or home-to-home deliveries. Depots are therefore distributed hubs or homes. Each delivery is assigned to a single drone. The authors formulate two Multi-Agent Path Finding (MAPF) problems in which they aim to minimise the maximum energy consumption among all drones while avoiding collisions and satisfying energy constraints. The service area is divided into small regions by a grid. Collisions can be of two types: node-collisions (that occur at a point in the area when more than two paths occupy the point) and edge-collisions (occurring between two adjacent points when two drones move from one point to the other in opposite directions). To solve the MAPF problem, the authors design the Negotiated Delivery Routing (NDR) algorithm, which returns a set of paths, one for each drone. NDR relies on the Negotiated Congestion (NC) Algorithm, which works by independently finding the shortest path of each pair

in every iteration, considering both the path distance and the collisions. The congestion cost and the slack ratio are iteratively updated to avoid collisions. The congestion cost is obtained based on the occupancy by paths and the congestion history of each node. In contrast, the slack ratio of a path is defined as the ratio between the path length and the maximum path length among all paths. For experimental results, the authors collected 30 hours of flight data, which included drone hovering, ascending, descending, horizontal flight, and random manoeuvres. All these factors are accounted for when modelling the energy consumption of drone flights, together with the parcels' weight. The NDR algorithm is compared with two classical methods for solving MAPF: the priority-based planning method (PR) and Conflict-Based Search (CBS). The results show that NDR has a higher delivery success rate, although it consumes more average maximum energy than CBS. The authors argue that CBS always finds the optimal solution as it searches for possible detours. However, to achieve this performance, the number of iterations for rerouting increases exponentially with the number of collisions, making CBS a non-feasible approach for dense networks. This is the only work that considers the problem of finding collision-free paths in a drone delivery context.

Shi et al. [16] formalize the multi-trip drone location routing problem with simultaneous pickup and delivery (MT-DLRP-SPD) problem. The MT-DLRP-SPD problem aims to decide how to dispatch drones in multiple central hospitals and plan their routes to cover the demands of epidemic prevention nodes for medical supplies delivery and sample pickup while minimizing costs and time. Differently from previously analysed works, the costs also include the decision to equip central hospitals with the necessary instrumentation and manpower to function as a launch site for drones, which comes with their availability in providing medical supplies and analysing the collected samples. The problem is formalized as a MIP and considers several candidate central hospitals, a number of epidemic prevention nodes, each identified with several deliveries and quantity to be picked up in kilograms, and a set of drones that are assumed to be homogeneous and able to perform multiple trips and to visit multiple nodes in one trip. The route planning also considers the power consumption limits of drones and is expressed in terms of the weight of the battery itself. To solve this problem, the authors implement the non-dominated sorting genetic algorithm II (NSGA-II), a classical multi-objective optimisation approach. The experimental results evaluate travel time and costs and do not include comparisons with other existing solutions. The authors also underline critical issues to take into account when planning the delivery of emergency supplies. Among those factors, the authors emphasize the importance of a fair resource distribution. Such issue is not considered in any other reviewed study.

c: MIXED

Mixing Single Depot and Support Stations, Arafat et al. [17] propose the Joint Routing and Charging Strategy (JRCS) for long-distance parcel delivery with multiple drones. The goal is to maximize the number of customers served within a minimum time. The authors model drones' flight duration depending on their payload and battery capacities and propose the Drone Delivery Route Optimization (DDRO) algorithm for minimizing the task time to deliver a set of parcels. This minimization is formalized as MILP. DDRO uses a clustering algorithm by splitting the delivery area into groups of customers with charging stations at their centroids. Within each cluster, a subroutine decides flight segments between the depot, customers, and charging stations. Then, for each drone, the MILP is solved. The performance evaluation shows that the proposed algorithm outperforms TSP and greedy algorithms in terms of average delivery time, travelled distance and success ratio.

Mixing Multiple Depot and Support Stations, Farajzadeh et al. [18] consider a scenario in which depots are third-party logistic providers, that provision drones and delivery products are provided by facilities. Drones depart from their depot, pick up parcels at facilities and serve customers before going back to the base depot. To reach long distances, drones are also allowed to stop at support stations where batteries can be recharged. The objective of the model is to minimize the delivery cost. The authors assume that the drones are homogeneous; however, this assumption is not realistic, since they are assumed to be provided by different vendors. The authors design a MILP formulation that is solved for a toy example (2 depots, 3 facilities, 8 customers, and 2 drones) using the GASMS software [66].

Despite their limitations, these works are examples of how diverse components can be mixed to envisage a complex delivery system that enables long-distance delivery.

2) HETEROGENEOUS FLEET

Table 5 summarises salient features for works addressing Multiple Drones Multiple deliveries with a heterogeneous fleet of drones.

a: SINGLE DEPOT

Torabbeigi et al. [19] consider the possibility of drone failures and propose two-stage stochastic scheduling to find multi-trip routes for a fleet of heterogeneous drones to meet the customers' demand. Drones depart from and return to the base depot, where they can be loaded up to their own payload capacity. No energy limitations are considered, and the energy consumption of drones is not modelled. The objective is to minimize the so-called Expected Loss of Demand (ELOD) due to drone failures, which are assumed to be exponentially distributed over time. ELOD is defined as the average weight of the lost demands as a failure occurs.

TABLE 5. Salient features for articles addressing delivery with multiple drones with heterogeneous fleet.

Reference	Ground components	Objectives	Energy limitations	Consumption model	Battery Management	Uncertainties	Methodology	Simulation data
Torabbeigi et al. (2018) [19]	SD	L	MFT			DF	MILP, H	SYD
Jung et al. (2021) [20]	SD	RU	BC	CM	REC		MILP	SYD, RD
Yuan et al. (2021) [21]	SD	T	MFT				CSP	SYD
Wen et al. (2022) [22]	SD	T, C	BC	CM[PW, DW]			CSP, H	SYD
Song et al. (2018) [23]	MD	nDP, D	MFT		REC		MILP, H	SYD, RD
Torabbeigi et al. (2019) [24]	MD	RU	BC	CM[PW]			MILP	SYD
Sawadsitang et al. (2019) [25]	MD	C, nDP, OT	MFT, MFD			DF	MILP	SYD, RD
Sawadsitang et al. (2021) [26]	MD	C	MFD, MDT			DF	MILP	SYD, RD
Pei et al. (2022) [27]	MD	T					H	SYD
Coelho et al. (2017) [28]	SS	D, T, RU	BC	CM	REC		MILP	SYD
Liu (2019) [29]	SS	T	BC	CM[Pw, DW]	REP		MILP	SYD
Bartolini et al. (2020) [30]	SS	SR	BC		REP		BKC	
Alkouz et al. (2020) [31]	SS	T	BC	CM	REC		DaaS	SYD, RD
Shahzaad et al. (2021) [32]	SS	T	BC	CM[PW, W]	REC	W, DF	DaaS	SYD
Ghelichi et al. (2021) [33]	SS	T	MFT	CM[PW, DW, BW, DR]	REC		MILP	SYD, RD
Tian et al. (2022) [34]	SS	T	BC	CM	REC	W	H	RD

The proposed approach is articulated as follows: the first stage finds a set of feasible solutions for a drone scheduling problem; then, for each route, the ELOD metric is computed; finally, the subset of feasible routes with minimum ELOD covering all customers is selected. The authors tested their approach only on one instance composed of 3 drones and eight delivery requests. The probability of drone failure follows an exponential distribution with the rate parameter λ set at 0.005. Results show that at the cost of a small increase in the total time that drones spend travelling (+2.27%), the reliability of drone scheduling is consistently increased (24.52%), meaning that the expected loss of demand is reduced. Although this paper pioneers the introduction of the reliability concept in drone delivery and incorporates payload factors, it does not account for meteorological conditions, which can significantly influence drone failures. Furthermore, the authors only test their solutions in a single experimental scenario.

Jung et al. [20] address a scheduling problem similar to the one described in [19]. In contrast to this work, Jung et al. consider remote islands with low accessibility instead of rooftops as delivery destinations. The proposed solution envisages the identification of a drone operation center (i.e. a depot) located on the mainland or on the larger/best-accessible island from which a fleet of homogeneous drones meets the delivery demands of remote islands. The authors proposed two mathematical models: an optimization model, which considers the wind speed and direction to cope with weather uncertainties; and an optimization model without climatic uncertainties as a baseline approach (which is similar to the one proposed in [8]). Parcels are located at the depot and are loaded one at a time on a drone which brings the parcel to the island and comes back to the depot.

Thus, a delivery task consists of completing a delivery and returning to the depot. Drones have limited battery capacity; the battery is assumed to be fully charged at the beginning of the schedule and can be recharged, if necessary, after every delivery task. Moreover, to prevent full battery discharge, the battery cannot be drained exceeding a minimum battery charge percentage. Another limitation of drones is that they can operate only a limited number of hours per day. The objective of the model is to remove the so-called “logistical dead zones” (i.e., islands that receive no parcels); hence, the optimization model intends on maximizing the minimum delivery ratio among the considered islands. Uncertainties considered in the robust model affect the flight time and the time for landing, unloading and taking off. To evaluate the proposed model, the authors conducted tests on both randomly generated and real data. The case study considers eight islands in the Go-Heung area in South Korea as remote islands and the Nohwa post office on Nohwado Island as the depot. Three sets of random data have been generated considering several wind scenarios: Beaufort wind scale two (6-11 km/h), three (12-19 km/h), and four (20-28 km/h). The results showed that as the wind speed increases, the first optimization model outperforms the model with no uncertainties. To generate a scenario based on real data, the weather data related to 19 days of July 2019 has been considered. Results showed that the first optimization performs better also with real data scenarios. However, the tests based on real data had few staples of Beaufort scales two and four. Thus results cannot be generalized. Finally, the authors conducted a cost analysis, showing that implementing the proposed system would increase cumulative savings over time, stating a saving of more than 6k\$ within five years of activity. The mathematical formulation introduced

in this paper accurately models several factors impacting the performance of drone delivery systems. The work proposes a convincing evaluation carrying out experiments based on real data. Nevertheless, the authors do not discuss the computational complexity of their proposed models. Instead, they provide the computation time of each day of experimentation, which is highly variable.

Yuan et al. [21] address the problem of minimizing the makespan of deliveries (i.e. the time at which the last parcel is delivered) using a fleet of heterogeneous drones in an urban environment. The delivery demand consists of parcels shipped from a base depot and delivered at stations deployed in the region of interest. Note that such stations are used only as delivery cabinets and do not provide drone battery replacement nor recharge. A drone schedule consists of a multi-trip route. Each trip starts at the base depot, reaches one or more stations to deliver parcels, and then returns to the depot. Trips are constrained by the drones' energy limitation expressed as the maximum flight time, and the number of parcels delivered during a tour is constrained by each drone's payload capacity. The authors formalized the problem as a MILP and proposed a genetic-based algorithm for solving the scheduling problem. The authors evaluated the proposed algorithm by comparing it with two general scheduling algorithms: tailored versions of the simulated-annealing-based algorithms proposed by Dorling et al. [7] and by Basbous et al. [67]. The algorithms are evaluated in relative percentage deviation, that is, for each algorithm and for each tested instance, the best result achieved compared to the best result obtained among the other two algorithms. The evaluation is carried out by varying several parameters, including the radius of the area covered by the drones, the number of stations, package weight and others. The results show that the proposed genetic algorithm outperforms the benchmarks in all considered scenarios, although the relative percentage deviation with respect to the simulated annealing algorithm is at most 0.57%, revealing that the proposed algorithm does not sensibly improve the benchmark solutions.

Wen et al. [22] address a routing problem based on heterogeneous multi-drone (HDDP) in which a large fuel-powered drone (mother drone or m-drone) carries a fleet of homogeneous small rechargeable multi-rotor drones, called *e-drones*, that deliver parcels. Drones can recharge at support stations (in this work, called airports). The authors propose a three-stage iterative optimization algorithm (IO-3S): in the first stage, the customers are clustered in K sub-regions (as many as *e-drones* in the system). One drone will serve each sub-region, and the total delivery demand is such that the overall payload weight does not exceed the *e-drone* maximum payload capacity. To define clusters, a modified fuzzy c-means function is adopted; in the second stage, the m-drone route is planned to start from the depot, reach all the clusters' centroids and finish at the depot. To solve this subproblem, the authors combined a variable neighbourhood

descent algorithm with a tabu list; finally, in the third stage, small drone routes are planned using dynamic programming. The route of the k -th drone starts at the centroid of the k -th cluster, reaches all customers in the cluster, and ends at a station. Note that it is assumed that each *e-drone* can reach the station with its battery capacity, and no recharging options are considered, nor is it considered the possibility of *e-drones* being launched again after the station has been reached. The three stages are mathematically formalized as a set of constraints to define the space of solutions of each sub-problem. The objective of the proposed algorithm is to minimize the overall delivery time and the total delivery cost. To find a near-optimal solution, a simulated annealing algorithm that includes these three stages has been proposed. The authors conducted extensive experiments on synthetic instances and a practical case. The proposed algorithm is compared against two algorithms for the truck-drone delivery problem (i.e. the same problem in which the support vehicle is not the mother drone but a ground vehicle) proposed in the literature. The synthetic instances are generated considering an increasing number of customers (i.e., 40, 60, 80, and 100) respectively deployed on an increasingly larger squared area of interest (i.e., 6 km², 8 km², 10 km², and 12 km²). Each instance has one parent drone and an increasing number of *e-drones* (i.e., 1, 2, 2, and 3). Tests performed on synthetic instances showed that the proposed algorithm finds better results concerning the best alternative proposed in the literature (respectively 30.42%, 28.12%, 26.12%, and 26.46% less time; and 21.79%, 17.64%, 14.64%, and 6% less cost). The real case test considers an instance with 80 customers in Changsha, China using 15 *e-drones*. Although the authors present promising results, the envisaged system might be far from being implementable since weak public acceptance might hinder it.

b: MULTIPLE DEPOTS

Song et al. [23] address the problem of maximizing the number of accomplished deliveries while minimizing the travel distance in a delivery system composed of a heterogeneous fleet of drones and provided with multiple depots. Delivery requests are modelled as tasks to be assigned to drones. Each task corresponds to a specific demand, characterized by the coordinates of the delivery point. Drones rely on multiple depots to recharge batteries and load parcels. The authors assume that the time needed to recharge the battery is constant and independent of the battery's state of charge. This assumption is more appropriate for battery replacement. A drone can serve multiple customers in each route that starts and ends in a depot. While performing the assigned tasks, each drone can carry a limited payload, whose effects on flight speed and consequent limitations on drones' flight time are considered. The authors conducted experiments to show that the proposed heuristic algorithm can obtain optimal or near-optimal solutions. To this aim, the heuristic solutions are compared with those obtained using the CPLEX solver.

Several scenarios have been generated, considering three depots and six drones. The authors studied the performance with various deliveries, i.e., 5, 10, 15, and 30. They generated ten scenarios for each considered demand volume. CPLEX solver could not solve all scenarios due to the memory limitations of the computing environment. However, for what concerns the instances that could be optimally solved, the heuristic algorithm always finds solutions for which the objective function's value is at most 3.7% less than the optimal value. These results are very promising, yet the paper only tests for the multi-function objective defined in their MILP. It would have been interesting to understand which of the two objective functions (namely, the number of covered tasks and the travelled distance) has the strongest impact, and to run comparisons with state-of-the-art solutions that optimize for one of the two objectives.

Torabbeigi et al. [24] present a model for minimizing the number of drones used to deliver parcels to multiple customers using a fleet of heterogeneous drones. A multi-trip route is computed for each drone; each trip reaches multiple customers and starts and ends at the drone's base depot. Drones might share the same base depot. Depots are used to store packages, and drones pick them up according to their payload capacity. Energy consumption is modelled as drone-specific and payload-dependent. Moreover, the authors consider preserving the minimum battery level to prevent battery full discharge so that a safe landing is ensured. The problem is first formalized as a MILP formulation, and then the solution approach consists of 1) preprocessing the instance to fix some variables to 0 before solving the model to reduce the search space; 2) computing a primal bound generation method providing an upper bound to the optimal solution; 3) several dual bound methods are proposed providing a lower bound to the optimal solution; 4) finally, if the primal and the dual bound solutions are equal, the optimal solution is obtained; otherwise, if the gap between them is below a certain threshold such threshold represents the accuracy of the solution found. The primal bound method consists of two steps: first, each customer is assigned to the nearest depot; then, the MILP problem is instantiated and solved for each depot to find the number of required drones and their flight paths. The dual-bound methods represent a relaxation of the original problem. The authors propose the following relaxations: 1) Lagrangian relaxation, which consists of moving some hard-to-solve constraints into the objective function introducing a penalty cost associated with such constraints; 2) Network configuration-based relaxation, which consists of computing the maximum clique in the customers' incompatibility graph (in which each edge represents two customers that cannot be served by the same drone); the size of the maximum clique represents a lower bound for the number of required drones; and 3) Drone weight capacity based relaxation, in which the problem is reduced to a Bin Packing Problem where drones are bins with a predefined capacity and deliveries are objects whose size corresponds to the customers' demand. The authors first

compared the results obtained with the proposed dual-bound methods without using the preprocessing algorithm. Six scenarios are considered, with two depots and 11 customers with growing demands and travel times. Results show that the Lagrangian relaxation algorithm obtains the best results but with the highest computational time, making it ineffective in practice. The dual bound method based on network configuration outperformed the other two methods, providing good bounds with low computational time. Then, the authors conducted a second set of experiments comparing the CPLEX solver and the bound methods; both of them with and without using the preprocessing algorithm. They considered three scenarios: 1) 1 depot and six customers, 2) 2 depots and ten customers, and 3) 1 depot and 14 customers. The results show that using the preprocessing algorithm reduces the computational time of the CPLEX solver and the bound methods. Moreover, the results show that in the considered scenarios, bounds methods generally find a solution as good as the optimal one with a lower computational time. A solid problem formulation is one of the strengths of this work. However, the paper lacks a sufficiently extensive evaluation. To assess the effectiveness of the proposed approach a larger variety of scenarios could be considered both synthetically generated and based on real data.

Sawadsitang et al. [25] consider a delivery network composed of multiple depots and a heterogeneous fleet of drones. The aim is to deliver to multiple customers, each with a demand that can be served by a drone or outsourced to a carrier. In the case of drone delivery, customers can express preferences on time windows for delivery. Drones have specific limits such as payload capacity, maximum distance per trip, maximum distance per day, average flight speed and period of flight availability. Each drone can fly from and return to only one depot. Packages can be transferred from an original depot to a new depot. The objective is to minimize the total delivery cost and the percentage of unsuccessful packages delivered due to drone failures while maximizing the reward of on-time delivery. To formalize this problem, the authors propose a three-stage stochastic model: in the first stage, customers are assigned to either one of the drones or outsourced to a carrier; in the second stage, it is evaluated if the drone can take off, in such case the drone route is defined starting from the drone depot accomplishing one delivery reaching a customer's location and returning to the drone depot, otherwise, all packages assigned to the drone are considered unsuccessful deliveries; the third stage evaluates if the drone breaks down during the delivery, then the package in the broken drone is considered as an unsuccessful delivery as well as the packages of the customers that should have been served by such drone after the breakdown occurs. The authors performed experiments based on real data provided by a Singapore logistics company. The dataset includes 40 customers, two depots and two drones. The authors present the Pareto frontier of the two proposed objectives, namely the minimization of the total cost and the minimization of the percentage of unsuccessfully delivered parcels. The results

shown in this paper provide interesting insights into how the total cost, the percentage of unsuccessful deliveries and the breakdown probability impact one another.

This last work has been extended by the same authors in Sawadsitang et al. [26]. This more recent work introduces the concept of shipper cooperation. The proposed framework is articulated in three components: “Shipper cooperation management”, which allows shippers to share drones and depots, creating a pool of resources; “Package assignment”, which is an extension of the model previously proposed; and the “Cost management”, that ensures that costs are distributed fairly among cooperative shippers. In accordance with the purpose of this article, we focus only on the “Package assignment” component. Given a coalition of cooperative shippers that share a resource pool composed of drones and depots, deliveries are carried out using the entire resource pool. Shippers can transfer packages from their depot to other depots in the pool. Deliveries can also be outsourced. The objective is to minimize the total delivery cost of a coalition of cooperative shippers, considering the probability of drone failure. The problem of assigning parcels to drones is formalized as a multistage stochastic optimization problem. Each drone cannot change depot and can deliver one parcel at each stage. Random drone breakdowns are considered to define the probability of failure that depends on drone history. Energy consumption is not modelled, but drone-specific limitations on kilometres per trip are considered. To evaluate the proposed solution, the authors considered four shippers, each with one drone and four depots dislocated in the area of interest. The authors performed simulations on synthetic data, considering 60 customers, and on real data, with 100 customers. Comparing a baseline approach in which no shipper cooperation is allowed, the authors show that a cost reduction is achieved (independently of considering or not the probability of drone breakdown). The total cost includes the cost of purchasing the drones, drones’ resources, and the cost of the potential relocation of packages. Despite the thorough validation of the solutions proposed in [25] and in [26], none of the works compare with other state-of-the-art solutions, making it hard to frame the contribution with respect to other existing works.

Pei et al. [27] formalize a MIP for food delivery assignments to fleets of drones. The MIP accounts for collision avoidance and aims to minimize the maximum total delivery time. The scheduling scheme is divided into stages, each composed of two phases: the dynamic phase, during which orders are received, and the static phase, which is the actual delivery of orders. Collisions can occur as drones use Computer Vision systems to identify the spot where to land, usually in the front house, and are usually provided with only one camera, which does not allow peripheral vision during landing and take off. Hence, drones are not provided with an autonomous system for avoiding crashes. The proposed MIP does not account for constraints such as power consumption, maximum flight time, and maximum flight distance, but it ensures that deliveries take place within

a promised time limit. To solve this problem, the authors propose two solutions. The first solution is a greedy heuristic in which, for each demand, the time to finalize the order of the current minimum loaded drone and the drone nearest to the pickup point are compared. The delivery order is assigned to the drone that can finalize it earlier. The second solution consists of relaxing the MIP and solving it with a branch-and-cut algorithm implemented in Gurobi. The two approaches are compared in terms of makespan and execution time relative percentage gaps in three main groups of simulations, small, medium and large scale, where the number of drones spans from 5 to 30, and the number of demands is two times the number of drones. The results show that the proposed algorithm and the branch-and-cut approach reach the same optimal solutions. However, for several instances in the medium and large-scale scenarios, Gurobi does not find the solution within the maximum execution time limit. As no comparison with previous state-of-the-art solutions for the minimization of the delivery time makespan is provided, it is hard to assess the results reported in the evaluation. A major strength of this paper, however, is the inclusion of a dynamic scenario where deliveries are received in an online mode. The demand dynamics is an important aspect that is typically neglected in existing prior work.

c: SUPPORT STATIONS

Coelho et al. [28] propose a drone delivery system in which the aerial space is divided into two layers: the upper layer, which covers high altitudes; and the lower layer, which covers low altitudes. The authors consider a heterogeneous fleet of drones and, depending on their size, drones can fly in the upper or lower layer. The upper layer is reserved for larger drones that are supposed to carry heavier loads, whereas the lower layer is reserved for smaller drones that can carry lighter loads. Drones rely on a transportation network of non-capacitated support stations where they can charge their batteries. Some support stations are also used by drones to exchange parcels among layers. Customers are scattered in arbitrary locations where drones have to deliver or collect parcels. Thus, the lower layer is reserved to transfer parcels between customers and support stations, whereas the upper layer is reserved to transfer parcels between support stations. Drones’ energy consumption is considered to be speed-dependent. The authors propose a model in which parcels cannot be exchanged between drones. Thus, to exchange parcels between layers, the model is instantiated multiple times, and the output of the solution of one layer is the input of the other layer. Therefore, the model aims to find the optimal solution to partially perform deliveries. The authors performed experiments considering six different types of drones for a total of 37 drones; 30 small drones with three different payload capacities and maximum speeds in the lower layer and 7 with three different payload capacities and maximum speeds in the upper layer. In their experiments, the routing model is instantiated only twice. The first instance considers the upper layer and 10 packages

that must be delivered at the support stations. The second instance considers the lower layer and the 10 packages delivered at the support stations from the upper layer and 15 additional packages. A total of five support stations are deployed for each layer. Three stations are at predefined positions and serve as exchanging points between layers. Two stations are only for recharging purposes and are deployed at random locations. Considering a multi-objective function, 64 combinations of weights are considered to generate MILP problems. The authors investigate the relationship between objectives, concluding that: 1) minimizing the last collection consequently minimizes the last delivery; 2) finishing the mission with more charged batteries is correlated to travelling longer distances; 3) increasing the number of used drones, there is a reduction in the mission completion time. In this paper, the authors propose a complex and sound mathematical MILP formulation for the drone delivery problem that they solve with a metaheuristic. A limitation of this work is the lack of complexity analysis and computational time of the proposed algorithm.

Liu [29] presents a real-time food-delivery system. In this work, a fleet of heterogeneous drones is responsible for picking up orders at shops (i.e., suppliers) when ready and delivering them to customers, which may require the parcel to be sent to an arbitrary location. When a drone picks up a certain parcel, it has to carry the parcel until it reaches the destination and the delivery is finalized. Each drone can carry a certain number of parcels depending on its capacity. The author includes the drone's weights and its payload in the energy consumption model. The model proposed by the author does not pursue a global optimum; in fact, it is proposed as an online algorithm that periodically instantiates a MILP model to provide an uninterrupted delivery service. The MILP model aims to plan drone routes and assign deliveries related to a short period of time. The objective function accounts for several delivery efficiency parameters. The author evaluated the long-run performance of the proposed online algorithm through a simulation study covering a 6-hour operation scenario. 353 delivery requests were randomly generated with an arrival rate of one arrival per minute, following a Poisson process. Four support stations and 50 drones were randomly deployed in the service area. The deployed drones are divided into three classes according to the size and payload capacity: 24 big fast drones with a capacity of 6 packages; 26 slower drones with different capacities, 13 with a capacity of 2 packages and 13 with a capacity of 3 packages. The results show that the computing times are tolerable for online deployment, with few orders experiencing long delays. Also, faster drones are used twice the slower ones, indicating that the work is not well balanced due to the prioritization of fast deliveries.

Bartolini et al. [30] present a federated framework named DRUBER that relies on a fleet of heterogeneous drones to provide real-time drone delivery services in an urban context. This approach's novelty lies in the fleet's composition and

cooperation. In fact, the fleet of drones and the support stations belong to private owners, making them available to end-users who want to send parcels from a source station to a destination station. To accomplish the delivery within the budget requirement, the authors allow parcel handover among drones. Having introduced cooperation among drones owned by people whose behavior cannot be assumed to be correct, trust issues have been addressed by leveraging a blockchain implementation. Furthermore, an evaluation of economic feasibility is provided. Not only do the authors present the first work addressing cooperation among heterogeneous drones owned by multiple stakeholders, but they also address trust issues arising from this type of cooperation.

Alkouz et al. [31] consider a single delivery request with a heavy payload or a large package that cannot be delivered with a single trip. The proposed solution consists of splitting the original package into smaller packages and assigning them to a fleet of drones that cooperate to deliver the package. The fleet can behave in two ways: static or dynamic. In the former case, the set of drones that perform the mission is determined at the source and travels together to the destination. In the latter case, the drones belonging to the fleet may change during the mission. Also, the fleet may split into sub-fleets and follow different paths. The static fleet has the advantage that all drones arrive at the destination simultaneously, whereas this is not guaranteed for the dynamic fleet. Indeed, in this latter case, it is necessary to constrain all the sub-fleets to arrive at the destination within a limited time window. However, dynamic behavior provides better resource utilization, reducing waiting times and congestion. In their experimentation, the authors considered up to 200 connected stations generated using an urban road network dataset from London. The authors synthetically generated 2000 requests with random sources and destinations with a maximum payload of 10 packages, each weighing a maximum of 5 kg. Thus, the authors consider a fleet of a maximum of 10 drones. For each request, they computed the shortest path from the source station to the destination station and grouped the obtained results by the number of stations involved in the shortest path averaging the results. They conducted several experiments and use the average delivery time as a performance metric. As their first experiment, their proposed algorithms are compared against a brute-force approach in which the fleet behaves statically and follows the path with the shortest delivery time and a Dijkstra-based algorithm in which the graph represents the transportation network. The cost of an edge is the summation of travel time, charging time, and waiting time. With this first experiment, the authors show that their proposed algorithms behave better than the Dijkstra algorithm, improving charging and waiting time. Then, they compared the static and dynamic behaviors, showing that the dynamic behavior in which the fleet split maximum into two sub-swarms significantly reduces the delivery time with respect to the static behavior. In another experiment, they show that allowing the fleet to split in more than two sub-swarms does not improve

(and sometimes degrades) the performance of the algorithm due to the dispersion of sub-swarms. Finally, the authors tested the static behavior with different strategies to select the next hop of the fleet. They show that looking ahead a few hops to choose which node to select as the next hop reduces delivery time but increases the execution time.

Shahzaad et al. [32] apply service-oriented approaches to model and compose drone services for delivery purposes. This framework envisages the existence of multiple deliveries, but the proposed algorithm considers a single one. In this work, the authors present an algorithm resilient to wind conditions. First, a heuristic solution to the multi-armed bandit tree exploration problem provides an offline composition of drone services. Such a solution is based on the definition of states that describe the model in terms of stations and the times at which the drone reaches the stations. The state exploration is guided by a look-ahead approach, meaning that the next-to-adjacent states are considered when making the state selection decision. Then, an online algorithm is proposed to adapt the service composition whenever a failure occurs. When a service fails, the proposed algorithm analyses the impact of the failure and locally recomposes a new plan that reaches a station in common with the initial plan. The authors performed tests with varying fleet sizes, 50 to 80, and the number of stations, 10 to 60, each equipped with five charging pads. Simulations have been performed considering 1500 scenarios with a delivery request with random source and destination stations. The authors considered delivery time and distance travelled as evaluation metrics. Firstly, the authors evaluated the performance of the offline algorithm that provides the initial plan. The proposed approach is compared with a brute-force approach, which provides the optimal solution, and with a different version of the proposed approach without the look-ahead selection method. The results show that the delivery time of the proposed approach is, on average closer to the optimal solution than the solution obtained without looking ahead in the state selection. Moreover, the traveled distance is higher in the brute force and the lookahead approach solutions with respect to the solutions provided by the search without the lookahead approach. The authors then evaluated the proposed online algorithm, comparing their approach with a brute force approach that consists in globally re-computing the plan upon failures. A plan's re-computation generally increases the travelled distance and delivery time. However, the authors observed that even if the failure rate increases, the proposed approach finds solutions close to the optimal brute-force solutions.

Ghelichi et al. [33] address the problem of routing a heterogeneous fleet of drones to deliver products from medical providers to clinics with a demand to be met. Given the set of possible trips for each drone, the objective is to find the schedule of trips needed to serve demands to minimize the total completion time while optimally locating charging stations. Each trip starts from a provider's location. The drone visits a sequence of charging stations, serves a

clinic, and returns to its corresponding provider. The energy consumption model is drone-specific and payload-dependent. Different drones may have different coverage ranges. The authors conducted experiments on synthetically generated data and data gathered in the Louisville area (USA). The synthetic data consists of 30 random instances with 20 urban providers, 30 candidate recharge platforms, and 25 rural clinics, each having a demand for two medical packages. On this dataset, two experiments were conducted. Firstly, the authors investigate the correlation between the number of batteries per drone, the number of medical packages that make up the payload, and the frequency of drone visits to charging platforms. Two scenarios were considered with the following constraints for drone setup: the first allowing two medical packages, one onboard battery, and at most two visits to charging platforms; the second allowing only one medical package, two onboard batteries, and at most one visit to charging platforms. The main findings were that in the first scenario, the increase in the number of installed platforms results in a decrease in the time that drones spend waiting to recharge their battery at stations. On the other hand, in the second scenario, fewer charging platforms have to be installed. Nevertheless, allowing only one package per drone, each clinic has to be visited twice, resulting in an increase in the overall time drones spend performing trips (more than doubled - 2.5). The Louisville case study consisted of 12 rural clinics and ten urban providers selected out of 28 available. Potential locations to install charging platforms were 20 between streetlights, traffic signals, and billboards. The authors first studied the impact of varying the maximum allowed number of drones per provider (1, 2, or 3). They considered the case in which only one visit to the charging platforms is allowed, but each drone has two batteries. They found that with this setting, installing only one platform allowing 1, 2 or 3 drones resulted in infeasible missions. By installing two charging platforms, feasible missions could be found, allowing 2 or 3 drones per provider. The authors concluded that the more drones per provider are allowed, the more the system performance depends on the number of installed charging platforms.

Jointly choosing the infrastructure location and the drone routes is crucial for optimizing charging platform locations based on demand. However, as the demand can change in terms of volume and location, it is impossible to position new charging platforms any time the drone routes are adjusted. Therefore, separating the decision-making process and factoring in the fluctuating demand may be more effective, ensuring the system remains scalable and adaptable. The scenario considered by the authors can be considered static, at least in the locations of the demand, and hence in the drone routes shape. Furthermore, the authors studied the impact of allowing longer operational daily time. They found out that it contributes to achieving better solutions in terms of overall time drones spend to perform delivery trips and delivery costs while increasing the time at which the last delivery trip is performed.

Tian et al. [34] propose algorithms for optimal charging time assignment and global optimum path planning in both static and dynamic scenarios for solar-powered UAVs (SPUs). They consider networks composed of a number of delivery stores, solar-powered drones and landing places, which work as charging stations and supplier facilities. The dynamism considers different time-dependent factors, including solar charging efficiency depending on the time of the day, limited landing space, changes in the city map, and emergencies due to weather changes. The authors design the charging time assignment (CTA) algorithm to find the optimal charging time assignment plan in any statically charging efficiency environment. This algorithm is integrated into the proposed global optimal algorithm (GOA) for finding optimal paths for solar-powered drones. The approach for static route planning and charging is extended to tackle dynamic environments. Their solutions are compared to other classical approaches for UAV delivery path planning, which are a distance-based and a weight-based Dijkstra algorithm that does not apply CTA, showing an improvement with respect to these approaches that spans from 6% to 80% in delivery time.

V. CHALLENGES

In this section, we review the aforementioned approaches highlighting the most common issues related to Drone-based Delivery Systems. Specifically, we want to discuss the following aspects: *What are the objectives of the proposed approaches? How are drone energy limitations modeled? What are the uncertainties, and how are they handled?* Our findings are summarized in Figure 5.

A. OBJECTIVES

The objective of a model has a significant impact on the performance of the system. Some actors, such as the service's customers, might want the parcels to be delivered fast; others, such as the companies providing delivery services, might want to reduce costs and optimally use their resources.

In the following, we describe several objectives that have been proposed by the analyzed works.

1) TIME

Completion time or *lateness* are objective functions that the final user is interested in minimizing.

In [2] and [5] an extension of the TSP problem is introduced to minimize the completion time which is defined as the sum of the times required to reach successive customers considering the ratio between the traversed distance and the drone's speed. The speed is influenced by the total payload carried by the drone while flying.

In [31] the authors compose address the realization of a "Swarm-based Drone-as-a-Service" to find a solution to the problem. Each service has a function that, given two nodes of the transportation network, provides the time needed to move from one node to the other; it considers the travel time, the charging time, and the waiting time due to recharging pad

congestion at support stations. The completion time is the sum of the times required by the services that compose the solution.

In [29] the authors formulate an optimization problem aiming at minimizing the lateness of orders' arrival. The lateness of an order is defined as the time elapsed from the moment the order is ready to be shipped to the time at which such order arrives at its destination.

The work in [33] the optimization problem decides if a drone completes a certain trip at a certain time slot and the corresponding decision variable is used in the objective function so as to minimize the sum of each trip's completion time.

In [32] a state search algorithm is proposed. Each state is associated with a timestamp that represents the arrival time at a certain node. The arrival time considers recharging times including the time the drone has to wait to recharge and the travel times.

The work in [11] formulates two optimizations for solving the trajectory planning and scheduling problems minimizing the delivery time. To solve these problems, the authors propose the hybrid genetic and simulated annealing algorithm and the UAV-Oriented MinMin algorithm, respectively.

The model proposed by [21] minimizes the *makespan* of deliveries which is defined as the time at which the last parcel is delivered.

2) COST

Delivery cost can be defined in terms of the number of drones to be deployed or as the energy consumed by the drones. Minimizing this metric is interesting from the perspective of the delivery service providers.

In [1] and [13], the problem of cost minimization is modeled through a graph where the costs are embedded in the edge cost function which changes over time according to wind observations. In [1], the edge cost is defined as the energy cost per unitary traversed distance given the wind speed and direction, multiplied by the length of such an edge. The objective of the proposed algorithms is to minimize the drone's energy consumption; thus, the proposed algorithm minimizes the sum of the cost of the edges composing the drone's route. The work in [13] minimizes the energy consumed by each drone by choosing the shortest path from the source to the destination as a drone's route. In this solution, the minimization of energy consumption results in increasing the percentage of successful deliveries. This is due to the re-computation upon edge costs changes of the drone routes that otherwise would have resulted to be energy infeasible.

In [26] the objective is to minimize several costs, including the initial cost to purchase the drones, the routing or outsourcing cost of drones, and the cost of the potential relocation of packages among depots. In case of drone failure, penalty costs are also considered.

The work in [18] minimizes is the total cost of delivery that is defined as the cost of drone routes and the cost of recharging batteries.

3) RESOURCE UTILIZATION

Companies providing delivery services can be interested in minimizing their resources. For example, in [24] the objective is to minimize the number of drones used to accomplish deliveries. However, optimizing resource utilization to make the system fair is also interesting from the customers' perspective. Indeed, in [20] the objective is to reduce/remove "logistical dead zones", namely areas in which the delivery service lacks. To do so, the objective function is defined as the lowest delivery ratio among all the customers (that in this work are remote islands with non-limited delivery demand). When maximized, this function reduces the possibility of islands not receiving any parcels.

4) DISTANCE

Some works focus on minimizing the maximum distance traveled by drones. The work in [12] proposes three mathematical formalizations for the Pickup to Delivery Drone Routing Problem, whose objective is to minimize the maximum distance traveled by a drone. The same objective is formulated in [4] in the case of a single drone carrying out multiple deliveries. With the assumption of constant drone speed, this objective is equivalent to minimizing the delivery time of each drone while serving all the demands. Nevertheless, drones' velocity can vary depending on other factors, including the weight of the parcels they transport and the weather. Other works focus on minimizing the overall traveled distance, e.g., the work proposed in [3].

5) OTHER

Both [9] and [10] propose a metric that expresses the satisfaction of customers, which is defined as the percentage of delivered demand, which has to meet a given threshold.

In [8] the objective is to maximize the number of parcels delivered within a given temporal horizon.

In [19] the objective is to minimize the expected loss of demand due to drone failure.

6) MULTIPLE

Some articles propose more complex formulations where multiple objectives are to be optimized simultaneously.

In [7], two alternative objectives are defined. The first is to minimize the completion time while keeping the cost below a given budget. The second is to minimize the cost of accomplishing the deliveries within a given temporal horizon. The cost is defined as the cost of the drones deployed and the cost of the energy required to perform all deliveries.

In [25] three alternative objectives are defined, although only two have been tested in the experiments. The first objective is to minimize the total cost of deliveries, defined as in [26]. The second objective is to minimize the percentage of unsuccessfully delivered packages. The third objective

proposed is to maximize the reward of on-time delivery. The authors define parameters to reward deliveries according to their arrival time. Customers are required to give an order of preferred time windows so that the reward given is proportional to the customers' preference.

In Shi et al. [16], the objective is to minimize travel time and costs simultaneously. Costs include the decision to use central hospitals as depots and drones' driving and energy costs.

In [28], a multi-criteria objective function is proposed. Seven goals are jointly minimized: the total distance traveled by drones; the total drones activity time; the number of used drones; the maximum speed reached by drones; the energy to be recharged; the makespan of collection, which is the last time-slot in which a package has been collected; and the makespan of delivery, which is the last time-slot in which a package has been delivered. These seven goals are combined linearly, and the coefficients are tunable to give adjust the weight of each component.

In [23] the objective is to maximize the number of delivered parcels while minimizing the total distance traveled by drones.

In [22], both the total delivery time and the total delivery cost are minimized. The delivery cost includes the fuel consumption of the parent drone and the energy consumed by small drones performing deliveries.

B. ENERGY LIMITATIONS

Drones rely on the limited power availability provided by their batteries. In planning routes, it is necessary to consider this limitation, which is modeled as a time constraint [8], [20], [21], [23], [33], in some studies also including a distance constraint [25], [26], or providing an explicit model for the drones' energy consumption [17].

The drone's energy consumption is influenced by several factors: the weight of the drone itself; the weight of the battery, which depends on its state of charge; the payload weight; and weather conditions such as wind speed and direction.

Some articles also propose models that prevent full battery discharge to enable safe landing [4], [8], [17], [20], [24], [28].

To allow long-distance coverage, batteries need to be swapped [7], [29] or recharged [8], [17], [20], [23], [28], [29], [31], [32], [33].

1) LIMIT ON DISTANCE/TIME A DRONE CAN TRAVEL

In [8], [19], and [33], the authors define the amount of daily available hours to accomplish deliveries. Such a limit is used as a time horizon to plan routes. In [20], [21], and [23] the authors define a drone-specific maximum travelling time.

In [25] and [26] drones are restricted to not exceed the traveling distance limit expressed as kilometers per day and the flying time limit.

The work in [22] specifies the maximum distance a small drone can cover from the location in which it is released by the mother drone and the station in which its route ends.

Of course, this affects the number of customers that can be served by a single drone.

2) ENERGY CONSUMPTION MODEL

Modeling the energy consumption of a drone in a delivery scenario faces additional challenges with respect to other applications. Some proposed models are simple: e.g., the work in [18] simply considers the energy consumed by a drone flying between two locations to update the battery state of charge and then make decisions on when it is necessary to recharge drone batteries, or the work in [28] models the consumption considering the energy consumed by the drone simply being turned on and the energy consumed by the drone moving. In this latter case, the first factor is a fixed, constant amount of energy consumption, and is computed by multiplying the time the drone is on by a given ratio indicating the energy consumption per hour. The second factor represents a variable consumption and depends on the speed of the drone, which is decided through the optimization problem. Hence, the computation of the energy consumed moving is the time spent flying multiplied by the given ratio indicating the speed-dependent energy consumption per hour.

In drone delivery, it is important to consider the effect of the payload weight on the drone's energy consumption. One simple way to account for this effect is the one proposed by [31] which assumes that a drone that is carrying 5 kg consumes 1% of its battery every 10 km of traveled distance.

Another way to account for the weight of the parcel is proposed by [24], in which the authors provide a linear function that depends on the parcel weight. They experimentally compute the battery discharge rate during flight. To obtain the coefficients of the linear function they perform linear regression on the data collected using a Phantom 4 Pro+ drone in hovering flight mode.

Energy consumption models might also consider some physical characteristics of the drone itself, such as its weight, size, etc.

In the formulation proposed by [29] the mass of objects is not measured in kg but in payload units representing the weight of a standard-sized meal order, sizes can be 1, 2, or 3 units. Consequently, the battery capacity is measured in what the authors defined *payload minutes*, which is the maximum amount of minutes that a drone can carry a certain payload. For instance, if the battery capacity is 600 payload minutes means that the drone can either carry a payload of size 1 for 600 minutes, or a payload size 2 for 300 minutes, or a payload of size 3 for 200 minutes. To compute the energy consumption, the weight of the drone and the size of the meal order are subtracted from the energy availability of the drone. The weight of the battery is not constant; it changes proportionally to its state of charge.

The work in [7] introduces a model that considers a non-constant battery weight. The authors start by considering the power consumed by a single-rotor helicopter for hovering, which depends on the thrust needed. To compute the thrust, both the weight W of the frame and the weight m of the battery

and the payload are considered. Then the authors derive an approximate equation for the power consumed by an n -rotor copter assuming that each rotor shares the total mass $W + m$, and finally, they apply linear regression to obtain the linear equation used to model the energy consumption.

In [33], the energy consumption model is taken from [68] and considers the mass of the frame and the cooling system of the drone, the battery and payload weights, the lift-to-drag ratio of the drone (which is the lift force divided by the drag force where the lift force is “the force that acts at a right angle to the direction of motion through the air” and the drag force is “the force that acts opposite to the direction of motion”) and the battery power transfer efficiency. The work in [17] considers the flight duration of a fully charged drone to be a function of its payload and its battery capacity. The energy consumption of a flying drone is modeled as the draining of energy to control flight stability by counteracting the forces of gravity, drag, and wind. In [15], Lee et al. propose an energy model that accounts for a drone hovering, ascending, descending, horizontal flight, random manoeuvres, and load weight. Another aspect to consider is the effect of weather conditions such as wind speed and direction. The wind can increase or decrease the energy consumed while flying. In fact, flying tailwind consumes less energy, and flying headwind consumes more energy compared to flying without wind.

Several articles also embed the effect of wind in their proposed consumption model. We recall that the ground speed of a drone is the combination of airspeed and wind speed whereas airspeed is the speed at which an aircraft is moving relative to the air it is flying in. By adjusting the airspeed, it is possible to correct the trajectory of the drone, compensating for the effects of the wind.

In [5], the drone velocity is expressed as a function of both the drone load and the effect of the wind. The authors highlight that the effect of the wind on the speed of the drone depends on the angle between the drone velocity and wind velocity vectors, which can result in a higher battery consumption when the wind is non-favorable, or it can speed up the drone otherwise.

In [1], the authors model the drone delivery system as a graph whose edges are weighted with the energy consumed to traverse them (i.e., to travel from one node to another). Given an edge, its length is multiplied by the unit energy cost, which is defined as the ratio between power consumption and the average ground speed. The power consumption depends on the thrust needed to move the drone in the desired direction. To compute the thrust, it considered the gravitational force and the total drag force. The gravitational force is computed considering the drone and the payload weight; and the drag force is computed considering the air density, the drone's relative airspeed, the area considering the radius of the rotor, and the drag coefficient. Also [13] embed energy consumption in the time-dependent edge cost function, but no detail on how such costs are computed is provided.

In [9] and [10], the authors define the amount of energy consumed flying between two locations considering the drag coefficient, the air density, the front surface, and the width of the drone, the weight of the drone and the carried payload and the airspeed.

When considering heterogeneous fleets, energy consumption might be drone-specific, as in [24], [28], and [29].

C. UNCERTAINTIES

Drones flying outdoors operate in a highly dynamic environment; therefore, uncertainties must be handled. In the selected articles two types of uncertainties are considered: wind effects and drone failures.

1) WIND

Some articles assume that the wind is constant during the period of interest/simulation [9], [10]. However, such an assumption is unlikely to meet real scenarios. A more realistic approach is to consider changing wind conditions. The literature proposes two general approaches: the first is to use measurements retrieved periodically from devices deployed in the area of interest to accurately represent the real wind conditions accurately [1], [13]; the second is to use weather forecast [20].

In [5], [17], and [32] only the effects on energy consumption are considered. However, wind can impact other factors, too.

In [9], the authors study the effect of wind on the power consumption and the delivery time of drones. They study the effect of wind in two scenarios, which differ depending on whether the airspeed or the ground speed is to be constant. Maintaining constant airspeed accounts only for wind direction, as the drone's direction has to be adjusted accordingly. Maintaining the ground speed constant also requires adjusting the airspeed to contrast headwinds.

In [10], the same authors apply the first strategy (ground speed constant) to provide some practical application of the proposed model. A direct consequence of the effects on energy consumption is that the range covered by a certain drone changes from the range covered without wind. This might cause the unreachability of some locations of interest. In this work, the authors apply their model to identify such unreachable locations. When addressing this challenge, it is important to ensure that drones return to safe locations such as depots or support stations. Changing wind conditions might cause the infeasibility of the initial plan. In this work, the authors propose a model to plan alternative routes to ensure drones are returning to the base depot.

In [20], a robust optimization model is proposed to deal with uncertainties arising from wind conditions. The weather forecast is used to define the budget of uncertainty and the uncertainty sets related to the flight time and to the time required to land, unload the parcel, and take off.

2) DRONE FAILURES

The work in [19] assumes that failures might happen while the drone is flying between two locations. Given a route, a drone

- What are the objectives of the proposed approaches?
 - Time
 - Delivery Time (minimize)
 - Makespan of delivery times (minimize)
 - Lateness (minimize)
 - Reward for on-time delivery (maximize)
 - Cost
 - Drone Purchase costs (minimize)
 - Drone Deployment (minimize)
 - Energy Consumed (minimize)
 - Penalty costs (minimize)
 - Recharging costs (minimize)
 - Loss due to drone failure (minimize)
 - Resource Utilization
 - Number of used drones (minimize)
 - Unserved zones (minimize)
 - Distance
 - Makespan of traveled distances (minimize)
 - Overall traveled distance (minimize)
 - Delivered parcels
 - Amount (maximize)
 - Percentage (customer satisfaction level) (maximize)
 - System Reliability
- How are drone energy limitations modeled?
 - Maximum Flight Time
 - Maximum Flight Distance
 - Battery Capacity
 - Energy consumption model, considering ...
 - Payload Weight
 - Drone Weight
 - Battery Weight
 - Drag Force
 - Wind
 - Other
- Which and how are the uncertainties handled?
 - Wind
 - Constant speed and direction
 - Changing speed and direction
 - Drone Failure
 - Flying
 - Taking off

FIGURE 5. Summary of challenges.

might fail in each segment that joins two successive locations. The authors calculate the probability of failure according to a travel time-dependent failure density function. The failure over a segment implies that all the parcels loaded onto the drone are lost.

In [25], two types of failure are considered: a drone might not be able to take off, and a drone might experience a breakdown. Whenever a particular drone experiences a failure, all deliveries that were supposed to be carried out by such drone are considered to be unsuccessful.

In [26], each drone's failure probability is defined based on historical data. The related drone cannot accomplish its scheduled deliveries when a breakdown happens. To reflect the additional costs due to drone failures, breakdowns are associated with penalties to compensate for undelivered parcels.

In [32], a drone experiences random failures with a certain frequency that spans from 10% to 50%, depending on the experimental setting.

Some works [15], [27] include a collision avoidance strategy to avoid possible drone crashes in a multi-drone, multi-depot scenario.

In Figure 5, we summarize the challenges addressed by the reviewed literature in drone delivery.

VI. CONCLUSION

This survey reviews several articles addressing the problem of route planning in Drone Delivery Systems.

We have organized the reviewed articles according to a novel classification that takes into account the system infrastructure, the fleet size and the volume of the delivery demand.

For each work, we have summarized the problem addressed, the solution proposed, and the obtained results.

Moreover, we have outlined how the solutions presented in the reviewed articles tackle challenges common to all the proposed DDSs, such as the objectives of the models and algorithms that have been proposed, the energy limitations of battery-powered drones, and whether any uncertainty is handled.

All the reviewed articles pose interesting challenges, both in terms of applied and theoretical research and provide complex mathematical formulations. However, none of the articles seek approximation algorithms with proven guarantees. Instead of envisaging ever-more complicated systems, future research efforts might be focused on finding approximation algorithms with proven bounds, which is key to formulating service-level agreements that are realistic and attainable.

Moreover, despite the growing research interest and the number of Drone Delivery Systems available in the academic literature, it seems there is a lack of real-field experiments, owing to the fact that flying a drone or a fleet of drones over urban areas requires patents and permissions from the local authorities. Some studies have, however, made significant strides by incorporating real-world data into their simulations. This has provided valuable insights into drone delivery systems' capabilities and limitations.

While scientific articles play a crucial role in advancing the theoretical understanding of route planning algorithms for drone delivery, the pursuit of real-field experimentation and testbed development seems to be predominantly driven by private companies (see appendix A). These companies, however, do not frequently publish articles detailing their algorithms and experimentation. Consequently, a significant amount of valuable knowledge and progress in the field may remain proprietary, hindering the dissemination of information and impeding the broader scientific community's understanding of the practical challenges and advancements in drone delivery.

In conclusion, drones represent a promising, environmentally friendly technology that can be exploited in many application scenarios. Delivery systems are moving toward this technology, and a few services and trials have been successfully implemented worldwide. Therefore, it is reasonable to believe that this will continue to be an essential area of applied research in the years to come.

APPENDIX A

DRONE DELIVERY IN INDUSTRY

In this section, we overview trials and applications proposed by the industrial sector from 2013 to the present.

A. AMAZON

The introduction of drones in delivery systems was first announced by Amazon in 2013. Their service was designed to reach locations in a range of 16 km carrying parcels of weight less than 2.25 kg. The project, called Amazon Prime Air, reached the milestone of accomplishing the first parcel delivery on December 7, 2016. A package was autonomously flown from Amazon's Cambridge depot to the house of a customer who lived in close proximity and the delivery was accomplished 13 minutes after the order was placed [69]. This first success initiated the first publicly available trial which was meant to serve customers located within several kilometres of Cambridge Amazon's depot. In 2019, Amazon announced that with the release of a new model of prime air drone, the system would have been operational in selected cities a few months later [70]. However, that was not the case and in the summer of 2022 a new sophisticated drone model, able to detect and avoid obstacles enabling operations without visual observers to allow deliveries over longer distances, has been released and employed in a new trial in Lockeford, California [71].

B. DHL

DHL claims to be the first company to integrate the so-called "parcelcopter" into its delivery chain. In December 2013 a small parcel was flown using a remotely piloted quadcopter crossing the Rhine River for a one-kilometre trip. Later in 2014, after some modifications and optimizations, the new parcelcopter was launched to deliver medications to the island of Juist, successfully crossing the North Sea. This time the drone was operated, for the first time in Europe, beyond the pilot's field of vision. Constantly monitored from ground stations, the drone autonomously flew for 12 kilometres at an altitude of 50 metres. The next generation of parcelcopter, a tilt-wing aircraft, was employed in a trial during the spring of 2016 to accomplish deliveries at relatively high altitudes (up to 1200 meters). Deliveries were accomplished fully autonomously, reducing the delivery time from a 30-minute road trip to 8 minutes. In 2018 a new generation of parcelcopters was developed to reach high speed and long-range distances [72]. In 2019 DHL started an international partnership with EHang to deploy a fully automated drone delivery service in urban areas in China [73].

C. ZIPLINE

Zipline is an American company founded in 2014 to develop a drone delivery system for medical supplies. In 2016, Zipline started its collaboration with the Rwandan government, launching the first drone delivery system used to deliver blood in remote areas. Drones were manufactured by the

company itself and were able to carry up to 1.5 kilograms flying up to 150 kilometres round trip. Parcels were delivered in 15 minutes on average. To avoid drones landing at drop-off locations, the company used a small parachute to ensure the parcel safely reached the ground [74]. In 2020 the company partnered with the government of Ghana to deliver Covid-19 test samples operating in densely populated urban areas [75]. In the same year, Zipline asked the FAA for permission to operate its drones in the US territory [76] and started delivering medical products in North Carolina, becoming, with two routes with round trips between 30 and 50 kilometres, the longest-range drone delivery service approved in the US [77]. Later in 2021, Zipline partnered with Walmart to launch an autonomous drone delivery service in Northwest Arkansas [78]. The company is continuously expanding its service area around the world.

D. WING

Zipline is not the first company to deliver in the US. In fact, Alphabet's Wing started delivering snacks and healthcare products in 2019 to residents in Christiansburg (Virginia) [79]. Wing drones can carry up to 1.2 kilograms of payload reaching the maximum speed of 20 m/s (104 km/h) and have a round trip limitation of 20 kilometres [80].

E. OTHER COMPANIES AND ORGANIZATIONS

In 2020 also UPS and CSV started delivering parcels in the US. Using Matternet's M2 drones the delivery service meets the medicines demand of a retirement community in Florida [81].

Still in the US, a stunning delivery happened on April 19, 2019. After three years of collaboration among universities and organizations in Maryland, a custom-made drone flew a kidney in 10 minutes covering 4.5 kilometres to a nearby hospital where surgeons successfully performed transplantation of the organ [82], [83]. This was not an isolated experiment, in 2022 another drone flew a lung for a 1.5-kilometre journey to Toronto General Hospital. The flight took just six minutes [84].

Also in Europe, several companies have explored drone delivery. In 2017, Cleveron, an Estonian company tested an autonomous drone service aimed at delivering cold drinks to customers at a public beach [85]. In the same year, in Iceland, the Israelian Flytrex started delivering commercial parcels with a remotely piloted modified DJI Matrice 600 drone [86]. In 2020, Manna, an Irish start-up, launched a six-week trial of a delivery service in Moneygall delivering parcels up to 4 kilograms to residents [87]. Meanwhile, in the United Kingdom, the Royal Mail started delivering parcels to the rural community on the Isle of Mull [88]. In July 2022, the British NHS started trials to deliver medicines for chemotherapy through Drone Delivery [89].

Also in Japan in 2021, JP Rakuten Logistics conducted a two-week trial in Chiba City. The drone trajectory was a 24-kilometre round trip from a parking lot to a high-rise

apartment complex. Deliveries were accomplished using a drone model jointly developed with Coretronic Intelligent Robotics Corporation, able to carry up to 7 kilograms [90].

APPENDIX B OTHER DELIVERY SYSTEMS

While this paper focuses on drone-only delivery, other delivery systems employ drones and automated robots for last-mile delivery in collaboration with other means of transport. As a corollary to our work, hereafter we provide a brief survey of research works focusing on this kind of systems.

- 1) **Hybrid truck and drone delivery.** Several works in the literature propose hybrid delivery systems where drones are employed for last-mile delivery together with traditional trucks (e.g., [91], [92], [93]). Some works focus on establishing the most convenient truck parking locations from which drones depart to enhance the overall service, e.g., [94]. Other works focus on cyclic drone flights, in which a drone launches and lands at the same node, e.g. [95] and [96]. Yang et al. [97] study the impact of traffic in the drone-truck delivery problem and propose a robust solution for the truck-and-drone tandem to maximize profit. The truck and drone delivery problem poses a number of challenges that, in addition to the ones related to flying drones, include the cooperation between drones and trucks and the problem of deploying and supplying trucks in areas that are possibly traffic-prone or hard to reach.
- 2) **Autonomous robot delivery.** The employment of autonomous robots for deliveries represents another scenario that aims to provide efficient delivery services while reducing the energy and traffic impact of traditional truck-only delivery. Although some challenges related to the autonomous robot delivery problem are similar to the ones posed by using drones, some major differences suggest that the two problems must be tackled separately. For instance, drones can travel at higher speeds and in direct Euclidean paths, whereas autonomous robots have multiple compartments, higher payload capacity, and extended operational range that enable multiple deliveries in a single trip [98]. Some works focus on robot-only routing to carry out deliveries in a limited time period, e.g., [99], [100], and [101]. Other works study how to integrate robots for last-mile delivery into a truck-equipped delivery system, e.g., [102], [103], and [104]. The work in [105] analyses three different models for food delivery: one based on drones, one based on robots, and a hybrid one. Their study shows that the latter is the most efficient in terms of waiting time.
- 3) **Public transportation.** The use of public transport for parcel delivery has become popular as an ecological alternative to truck delivery, e.g., [106] and [107]. Recently, some solutions have been developed for

integrating public transportation and drones for parcel delivery. The authors of [108] and [109] propose an exact algorithm for a single drone that has to carry out multiple deliveries. The drone exploits a public train as a depot, which stores the parcels and works as a charging station for the drone. The work in [110] formalizes the problem of minimizing the return instant to a depot in a network of multiple drones, multiple depots and multiple vehicles and proposes an algorithm for solving this problem. In [40], the same authors formalize the reliable drone path planning problem, whose objective is to maximize the probability of delivering parcels to the customer on time. The problem includes random variables representing various traffic factors, such as congestion and delays. The work in [111] uses branch and bound to solve the problem of minimizing the delivery time for a single drone that delivers parcels exploiting as a depot a larger vehicle, which may be a ship or an aeroplane.

Delivery systems based on the collaboration between drones and means of public transportation pose different challenges: departing and landing on a moving vehicle is generally more difficult for autonomous drones; hence, such systems are more vulnerable to drone failures. Furthermore, the activity of the involved drones has to be adapted to the routes and the timetables of the means of transport which are involved in the system. Such a system also has to cope with architectural barriers, such as tunnels, and possibly with delays and rescheduling of the public transport's timetable.

APPENDIX C DRONE DELIVERY CHALLENGES

In addition to task assignment and trajectory planning for drone delivery systems, other challenges related to the design and implementation must be tackled in the realization of a real DDS. We present the most peculiar ones in this appendix.

According to [112], the major technological challenge is the flight-time duration limit. The work underlines that minimizing the impact of wind direction could mitigate energy-related issues.

Data communication is another major challenge in drone applications; hence, also in drone delivery [113]. For instance, the work in [114] proposes a low computational complexity algorithm to improve communication and connectivity in drone-based applications. The proposed algorithm is designed to solve issues arising from the introduction of adaptive antenna arrays, such as the radiation patterns not accounting for drones' mobility. Thus, the authors provide an algorithm that continuously estimates the direction from which the signal is arriving at the antenna array, allowing the radiation pattern to be adjusted in real-time to the current location of the drone.

Addressing a similar problem, the work in [115] aims at maximizing the area covered by communications.

Safety and security are other challenges that need to be addressed in the design of a DDS. In fact, cyber and physical threats, invasion of privacy, flying into restricted airspace, colliding, and causing property damage or harm to people are some of the possible issues related to security and safety. Thus there is the necessity for accountability of activities, logging protocols, and digital signature algorithms [116], [117].

Another complex challenge is the integration of drones into existing airspace systems so as to prevent accidents with other aircraft and obstacles [118].

Moreover, increasing the number of aircraft is necessary to regulate and manage the resulting aerial traffic to avoid congestion [119].

Finally, another challenge is related to the actual sustainability of drone technology. On the one hand, drones can reduce CO_2 emissions by reducing traffic congestion and fossil fuel consumption. On the other hand, drone delivery systems inherit the limitation of battery-powered systems. Namely, the source of energy used to charge the batteries can be more or less clean/green and battery production and disposal are still rather pollutant and concerning processes (due to raw material extraction, water consumption, use of hazardous chemicals, and human rights violation) [120].

In summary, while the heart of drone delivery systems lies in task assignment and trajectory planning, addressing the multifaceted challenges of communication, security, and sustainability is indispensable in ensuring their safe, effective, and scalable deployment in real-world contexts.

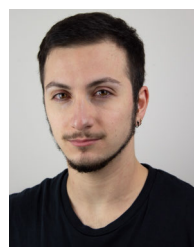
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GIULIO ATTENNI (Graduate Student Member, IEEE) received the bachelor's and master's degrees in computer science from the Sapienza University of Rome, Italy, in 2019 and 2022, respectively, where he is currently pursuing the Ph.D. degree in computer science. His research interests include sustainable mobility, green networking, and algorithmic fairness. His current research activities focus on route planning and task assignment for networks of aerial drones.



VIVIANA ARRIGONI (Member, IEEE) received the Ph.D. degree in computer science from the Sapienza University of Rome, Italy. She is currently a Research Fellow with the Department of Computer Science, Sapienza University of Rome. She has coauthored several research papers in networking, information theory, and computational linear algebra. She served as a TPC member for various international conferences.



NOVELLA BARTOLINI (Senior Member, IEEE) received the degree (Hons.), in 1997, and the Ph.D. degree in computer engineering from the University of Rome, Italy, in 2001. She was a Visiting Professor with Penn State University, from 2014 to 2017. She is currently a Full Professor with the Sapienza University of Rome. Her research interests include wireless networks and network management. She was the program chair and a program committee member of several

international conferences. She serves on the editorial board for IEEE/ACM TRANSACTIONS ON NETWORKING.



GAIA MASELLI (Member, IEEE) received the Ph.D. degree in computer science from the University of Pisa, Italy. She is currently an Associate Professor with the Department of Computer Science, Sapienza University of Rome, Italy. She participated in many European Community research projects. Her current research interests include the design and implementation aspects of mobile networks and wireless communications systems, with a particular focus on drones, back-scattering networks, and the Internet of Things, and the design and performance evaluation of networking protocols for RFID systems. She serves as a member of the TPC for several international conferences. She is an Associate Editor of *Computer Communications* (Elsevier). She serves as a Reviewer for several journals, such as IEEE TRANSACTIONS ON MOBILE COMPUTING and IEEE/ACM TRANSACTIONS ON NETWORKING.

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